

SECTION B: HEADINGS AND TEXT TAGS

2.1 <h1 to h6>

1. Tag Name

- **Tag Name:** <h1 to h6>
- **Opening & Closing Form:** <h1>...</h1> to <h6>...</h6>
- **Tag Type:** Paired / Container Tags

2. Definition

The <h1> through <h6> elements represent six levels of section headings in HTML documents. <h1> represents the highest section level (most important) and <h6> represents the lowest level.

3. Purpose / Function

Used to structure document hierarchy, organize text content logically, improve readability, and signal content importance to search engine indexing crawlers and accessibility screen readers.

4. Syntax

```
<h1>Heading Level 1 (Main Title)</h1>
<h2>Heading Level 2</h2>
<h3>Heading Level 3</h3>
<h4>Heading Level 4</h4>
<h5>Heading Level 5</h5>
<h6>Heading Level 6 (Sub-heading)</h6>
```

5. Basic Example

```
<h1>Computer Science Engineering</h1>
<h2>Semester 3 Syllabus</h2>
<h3>Web Technologies</h3>
```

6. Output

Renders headings with descending font size and font weight by default browser styles (<h1> largest bold, <h6> smallest bold).

7. Attributes

Attribute	Description	Example	Required/Optional
align	Sets text alignment (left , center , right , justify).	align="center"	Deprecated (Use CSS)
Global Attributes	Supports id , class , style , title , lang .	id="unit-1"	Optional

8. Attribute Example

```
<h1 id="main-heading" class="text-primary text-center">Data Structures</h1>
<h2 class="section-title">Binary Search Trees</h2>
```

9. Real-World Example

- **Use Case:** Used across all articles, blogs, documentation sites (e.g. MDN, Wikipedia) to establish semantic content outline.
- **Code:**

```
<article>
  <h1>Understanding Operating Systems</h1>
  <h2>1. Process Management</h2>
  <h3>1.1 CPU Scheduling Algorithms</h3>
</article>
```

10. Important Notes

Only use ONE `<h1>` per page representing the page main topic. Never skip heading levels (e.g., do not jump from `<h1>` directly to `<h3>`). Avoid using headings solely to resize text; use CSS for visual sizing.

11. Related Tags

Related to `<header>`, `<hgroup>`, and structural sectioning tags like `<section>` and `<article>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tags. `align` attribute is deprecated in HTML5.

2.2 `<p>`

1. Tag Name

- **Tag Name:** `<p>`
- **Opening & Closing Form:** `<p>...</p>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<p>` tag defines a paragraph of text. Browsers automatically add vertical margin (blank space) before and after a paragraph element.

3. Purpose / Function

Groups related sentences into distinct block paragraphs to organize written textual content cleanly.

4. Syntax

```
<p>Paragraph text content goes here...</p>
```

5. Basic Example

```
<p>HTML stands for HyperText Markup Language. It is the standard markup language for documents d
```

6. Output

Displays a block of body text separated from adjacent content by top and bottom margins.

7. Attributes

Attribute	Description	Example	Required/Optional
<code>align</code>	Specifies alignment of text inside paragraph.	<code>align="justify"</code>	Deprecated (Use CSS)
Global Attributes	Supports <code>id</code> , <code>class</code> , <code>style</code> , <code>title</code> .	<code>class="lead"</code>	Optional

8. Attribute Example

```
<p class="lead-text" style="line-height: 1.6;">This paragraph introduces the assignment topic wi
```

9. Real-World Example

- **Use Case:** Used everywhere on the web for news stories, blog posts, documentation, and product descriptions.
- **Code:**

```
<section>
  <p>Our college was established in 1995 with a vision to provide quality technical education.</p>
  <p>We offer undergraduate programs in CSE, ECE, and Mechanical Engineering.</p>
</section>
```

10. Important Notes

`<p>` is a block-level element. It cannot contain other block-level elements such as `<div>`, `<table>`, or `<h1>` - `<h6>`. Browsers automatically close a `<p>` tag when encountering another block element.

11. Related Tags

Related to `<br>`, `<hr>`, `<pre>`, and `<span>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. `align` attribute is deprecated.

2.3 `<br>`

1. Tag Name

- **Tag Name:** `<br>`
- **Opening & Closing Form:** `<br>` (No closing tag)
- **Tag Type:** Empty / Void Tag

2. Definition

The `<br>` tag produces a line break (newline) in text. It moves subsequent content directly to the next line without creating a new paragraph margin.

3. Purpose / Function

Forces a line break where line breaks are significant, such as in poems, postal addresses, or lyrics.

4. Syntax

```
Line 1 text<br>Line 2 text
```

5. Basic Example

```
<p>Department of Computer Science<br>Engineering Block A<br>VTU Campus, Belagavi</p>
```

6. Output

Renders the address on three separate consecutive lines without paragraph gaps between them.

7. Attributes

Attribute	Description	Example	Required/Optional
<code>clear</code>	Indicates where next line should begin after floating elements.	<code>clear="all"</code>	Deprecated (Use CSS clear)

8. Attribute Example

```
<span>First Line</span><br><span>Second Line</span>
```

9. Real-World Example

- **Use Case:** Used in contact address display cards, poem stanzas, and form receipt displays.
- **Code:**

```
<address>
  John Doe<br>
  123 College Road<br>
  New Delhi, India 110001
</address>
```

10. Important Notes

Do NOT use `<br>` to create visual vertical spacing between paragraphs; use CSS margins instead for accessibility and maintainability.

11. Related Tags

Related to `<p>`, `<address>`, and `<pre>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. `clear` attribute is deprecated.

2.4 `<hr>`

1. Tag Name

- **Tag Name:** `<hr>`

- **Opening & Closing Form:** `<hr>` (No closing tag)
- **Tag Type:** Empty / Void Tag

2. Definition

The `<hr>` tag represents a thematic break between paragraph-level elements in an HTML page (e.g., a shift of topic in a story or scene transition).

3. Purpose / Function

Visually renders as a horizontal rule/divider line separating different sections or topics.

4. Syntax

```
<hr>
```

5. Basic Example

```
<h3>Section 1: Introduction</h3>
<p>Content of section 1.</p>
<hr>
<h3>Section 2: Methodology</h3>
<p>Content of section 2.</p>
```

6. Output

Renders a thin horizontal dividing line across the width of the parent container separating Section 1 from Section 2.

7. Attributes

Attribute	Description	Example	Required/Optional
<code>align</code>	Sets alignment (<code>left</code> , <code>center</code> , <code>right</code>).	<code>align="center"</code>	Deprecated
<code>color</code>	Sets line color.	<code>color="blue"</code>	Deprecated
<code>size</code>	Sets line thickness in pixels.	<code>size="4"</code>	Deprecated
<code>width</code>	Sets line width in px or %.	<code>width="80%"</code>	Deprecated

8. Attribute Example

```
<!-- HTML5 Standard formatting via CSS -->
<hr style="border: 0; height: 2px; background: #cbd5e1; margin: 20px 0;">
```

9. Real-World Example

- **Use Case:** Used between article sections, modal dialog footers, and blog comment dividers.
- **Code:**

```
<article>
  <p>End of Chapter 1 summary.</p>
  <hr>
  <p>Chapter 2 begins here.</p>
</article>
```

10. Important Notes

In HTML5, `<hr>` is defined semantically as a thematic break, not just a purely visual decorative line.

11. Related Tags

Related to `<section>`, `<article>`, and CSS border styling.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Presentation attributes (`color`, `size`, `width`, `align`, `noshade`) are obsolete.

2.5 `<pre>`

1. Tag Name

- **Tag Name:** `<pre>`
- **Opening & Closing Form:** `<pre>...</pre>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<pre>` tag defines preformatted text. Text inside `<pre>` is displayed in a fixed-width monospace font and preserves both whitespace spaces and line breaks exactly as typed in the source code.

3. Purpose / Function

Used to present code listings, ASCII art, tabular text, or formatted computer output without HTML space collapsing.

4. Syntax

```
<pre>
  Preserved line 1
    Indented line 2
</pre>
```

5. Basic Example

```
<pre>
#include <iostream>
using namespace std;
int main() {
    cout << "Hello World!";
    return 0;
}
</pre>
```

6. Output

Renders C++ source code with exact spacing and tab indents intact using a monospace font (Courier/Consolas).

7. Attributes

Attribute	Description	Example	Required/Optional
<code>width</code>	Specifies maximum characters per line.	<code>width="80"</code>	Deprecated
Global Attributes	Supports <code>class</code> , <code>id</code> , <code>style</code> .	<code>class="code-block"</code>	Optional

8. Attribute Example

```
<pre class="bg-dark text-light p-3">
  SELECT * FROM Students WHERE gpa > 3.5;
</pre>
```

9. Real-World Example

- **Use Case:** Used extensively on technical sites like StackOverflow, GitHub, and tutorial platforms to display multi-line source code.
- **Code:**

```
<pre><code>
function add(a, b) {
  return a + b;
}
</code></pre>
```

10. Important Notes

Combine `<pre>` with `<code>` for code snippets. Reserved HTML characters like `<` and `>` inside `<pre>` must still be escaped as `<` and `>`.

11. Related Tags

Related to `<code>`, `<kbd>`, `<samp>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. `width` attribute is obsolete.

2.6 `<blockquote>`

1. Tag Name

- **Tag Name:** `<blockquote>`
- **Opening & Closing Form:** `<blockquote>...</blockquote>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<blockquote>` element indicates that the enclosed text is an extended quotation taken from another external source.

3. Purpose / Function

Renders indented block quotation text, attributing cited material in academic papers, blogs, and news reports.

4. Syntax

```
<blockquote cite="URL">  
  Quoted text content  
</blockquote>
```

5. Basic Example

```
<blockquote cite="https://www.w3.org/TR/html52/">  
  HTML5 is the 5th major version of the core language of the World Wide Web.  
</blockquote>
```

6. Output

Displays the quoted paragraph indented from both left and right margins.

7. Attributes

Attribute	Description	Example	Required/Optional
<code>cite</code>	Specifies URL pointing to the original source document.	<code>cite="https://quote-source.com"</code>	Optional

8. Attribute Example

```
<blockquote cite="https://en.wikipedia.org/wiki/Alan_Turing">  
  <p>Sometimes it is the people no one imagines anything of who do the things that no one can im  
</blockquote>
```

9. Real-World Example

- **Use Case:** Used in news articles, reviews, academic websites, and quote cards.
- **Code:**

```
<figure>  
  <blockquote cite="https://quotes.com/steve-jobs">  
    <p>Design is not just what it looks like and feels like. Design is how it works.</p>  
  </blockquote>  
  <figcaption>— Steve Jobs</figcaption>  
</figure>
```

10. Important Notes

Use `<blockquote>` for multi-line block quotes. For short inline quotes inside a paragraph, use `<q>`.

11. Related Tags

Related to `<q>`, `<cite>`, `<figure>`, `<figcaption>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Not deprecated.

2.7 <q>

1. Tag Name

- **Tag Name:** <q>
- **Opening & Closing Form:** <q>...</q>
- **Tag Type:** Paired / Container Tag

2. Definition

The <q> tag defines a short inline quotation. Most web browsers automatically surround text in <q> tags with quotation marks (e.g. "...")

3. Purpose / Function

Used for short, inline quotes that fit inside a running paragraph sentence.

4. Syntax

```
<p>Text <q cite="URL">Inline quotation</q> continued text.</p>
```

5. Basic Example

```
<p>As Neil Armstrong famously said, <q>That's one small step for man, one giant leap for mankind</q>
```

6. Output

Renders: As Neil Armstrong famously said, "That's one small step for man, one giant leap for mankind."

7. Attributes

Attribute	Description	Example	Required/Optional
cite	Specifies URL location of original quote source.	cite="https://nasa.gov"	Optional

8. Attribute Example

```
<p>The professor reminded us, <q cite="https://syllabus.edu">Assignments are due at midnight.</q>
```

9. Real-World Example

- **Use Case:** Used in blog posts, news stories, and online literary essays.
- **Code:**

```
<p>Sir Tim Berners-Lee stated that <q cite="https://w3.org">The Web as I envisaged it, we haven't</q>
```

10. Important Notes

Do not type literal quotation marks inside <q> tags; browsers inject locale-appropriate quotation marks automatically.

11. Related Tags

Related to `<blockquote>` and `<cite>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Not deprecated.

2.8 `<abbr>`

1. Tag Name

- **Tag Name:** `<abbr>`
- **Opening & Closing Form:** `<abbr>...</abbr>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<abbr>` element represents an abbreviation or acronym (e.g., HTML, CPU, NASA, IEEE).

3. Purpose / Function

Provides the full expansion of an abbreviation via the `title` attribute, showing a tooltip when hovered by users.

4. Syntax

```
<abbr title="Full Expansion Text">Abbreviation</abbr>
```

5. Basic Example

```
<p>We are learning <abbr title="HyperText Markup Language">HTML</abbr> in CSE department.</p>
```

6. Output

Renders "HTML" usually with a dotted underline. Hovering over it displays tooltip "HyperText Markup Language".

7. Attributes

Attribute	Description	Example	Required/Optional
<code>title</code>	Provides full phrase/definition of the abbreviation.	<code>title="Central Processing Unit"</code>	Recommended

8. Attribute Example

```
<p>The <abbr title="Institute of Electrical and Electronics Engineers">IEEE</abbr> standard is f
```

9. Real-World Example

- **Use Case:** Used in technical documentations, medical portals, academic papers, and government portals.
- **Code:**

```
<p>Contact the <abbr title="Doctor of Philosophy">PhD</abbr> department for research admissions.
```

10. Important Notes

Replaces the obsolete `<acronym>` tag. Crucial for screen readers to spell out acronyms correctly.

11. Related Tags

Related to `<dfn>` and `<cite>`. Replaces `<acronym>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Not deprecated.

2.9 `<address>`

1. Tag Name

- **Tag Name:** `<address>`
- **Opening & Closing Form:** `<address>...</address>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<address>` HTML element indicates that the enclosed HTML provides contact information for a person, department, or organization.

3. Purpose / Function

Encapsulates email addresses, physical locations, phone numbers, social media handles, or website authors.

4. Syntax

```
<address>  
  Contact details...  
</address>
```

5. Basic Example

```
<address>  
  Written by <a href="mailto:author@college.edu">Aryansh Sharma</a>.<br>  
  Visit us at: College Campus, Block 4<br>  
  New Delhi, India  
</address>
```

6. Output

Renders enclosed text in italic font by default, separated as a block element.

7. Attributes

Attribute	Description	Example	Required/Optional
Global Attributes	Supports <code>id</code> , <code>class</code> , <code>style</code> .	<code>class="contact-info"</code>	Optional

8. Attribute Example

```
<address class="footer-contact">  
  Email: <a href="mailto:info@univ.edu">info@univ.edu</a>  
</address>
```

9. Real-World Example

- **Use Case:** Used in page footers (`<footer>`), article author contact blocks, and company About Us pages.
- **Code:**

```
<footer>  
  <address>  
    Contact Dean of Academics: <a href="tel:+911123456789">+91 11 2345 6789</a>  
  </address>  
</footer>
```

10. Important Notes

Should only contain contact information relevant to its surrounding context (e.g. article author or site owner), not arbitrary physical addresses.

11. Related Tags

Related to `<footer>` and `<a>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Not deprecated.

2.10 `<cite>`

1. Tag Name

- **Tag Name:** `<cite>`
- **Opening & Closing Form:** `<cite>...</cite>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<cite>` tag marks a reference to a cited creative work (e.g., a book, paper, essay, poem, song, movie, artwork, website).

3. Purpose / Function

Provides semantic citation of title references in academic writing and literature citations.

4. Syntax

```
<p>Text about work <cite>Title of Work</cite>.</p>
```

5. Basic Example

```
<p>More details can be found in <cite>Introduction to Algorithms (CLRS)</cite>.</p>
```

6. Output

Renders the title "Introduction to Algorithms (CLRS)" in italicized font.

7. Attributes

Attribute	Description	Example	Required/Optional
Global Attributes	Standard global attributes.	<code>class="book-title"</code>	Optional

8. Attribute Example

```
<p>Refer to chapter 3 of <cite class="source-ref">Modern Operating Systems</cite> by Tanenbaum.</p>
```

9. Real-World Example

- **Use Case:** Used on online research repositories, university library catalogs, and Wikipedia references.
- **Code:**

```
<p>As described in <cite>The C Programming Language</cite> by Kernighan and Ritchie...</p>
```

10. Important Notes

Must contain the title of a work, not the name of a person.

11. Related Tags

Related to `blockquote`, `q`, `abbr`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Not deprecated.

2.11 `<code>`

1. Tag Name

- **Tag Name:** `<code>`
- **Opening & Closing Form:** `<code>...</code>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<code>` tag displays its contents formatted in a manner intended to indicate that the text is a short fragment of computer code.

3. Purpose / Function

Formats inline computer programming terms, variable names, function signatures, or HTML tag names in a monospace font.

4. Syntax

```
<p>Use the <code>functionName()</code> to compute values.</p>
```

5. Basic Example

```
<p>In Python, use the <code>print()</code> function to output text to the console.</p>
```

6. Output

Renders `print()` in a monospace font (e.g., Courier) embedded inline inside paragraph text.

7. Attributes

Attribute	Description	Example	Required/Optional
Global Attributes	Supports <code>class</code> for syntax highlighter integration.	<code>class="language-cpp"</code>	Optional

8. Attribute Example

```
<code class="language-javascript">const count = 10;</code>
```

9. Real-World Example

- **Use Case:** Used throughout programming documentation sites like MDN, GeeksforGeeks, and W3Schools.
- **Code:**

```
<p>To check file status in Git, type <code>git status</code> into terminal.</p>
```

10. Important Notes

For multi-line block code snippets, enclose `<code>` inside a `<pre>` element.

11. Related Tags

Related to `<pre>`, `<kbd>`, `<samp>`, `<var>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Not deprecated.

2.12 `<kbd>`

1. Tag Name

- **Tag Name:** `<kbd>`
- **Opening & Closing Form:** `<kbd>...</kbd>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<kbd>` tag represents user input, typically keyboard key entries, voice commands, or manual input instructions.

3. Purpose / Function

Styles text to visually represent keyboard shortcuts or user input keys.

4. Syntax

```
<p>Press <kbd>Key</kbd> to execute.</p>
```

5. Basic Example

```
<p>Press <kbd>Ctrl</kbd> + <kbd>C</kbd> to copy selected text.</p>
```

6. Output

Renders "Ctrl" and "C" in monospace style (often styled with CSS to look like physical keyboard key caps).

7. Attributes

Attribute	Description	Example	Required/Optional
Global Attributes	Standard global attributes.	<code>class="key-btn"</code>	Optional

8. Attribute Example

```
<p>Save file using <kbd class="keycap">Ctrl</kbd> + <kbd class="keycap">S</kbd>.</p>
```

9. Real-World Example

- **Use Case:** Used in software user manuals, OS shortcut cheatsheets, and web IDE tutorials.
- **Code:**

```
<p>To open Developer Tools in Chrome, press <kbd>F12</kbd> or <kbd>Ctrl</kbd> + <kbd>Shift</kbd></p>
```

10. Important Notes

Can be nested (e.g. `<kbd><kbd>Ctrl</kbd> + <kbd>V</kbd></kbd>`) to indicate combined shortcut gestures.

11. Related Tags

Related to `<code>` , `<samp>` , `<var>` .

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Not deprecated.

2.13 `<samp>`

1. Tag Name

- **Tag Name:** `<samp>`
- **Opening & Closing Form:** `<samp>...</samp>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<samp>` element represents sample output from a computer program or system.

3. Purpose / Function

Used to display terminal messages, system error codes, or console outputs in documentation.

4. Syntax

```
<p>Server returned: <samp>Sample output</samp></p>
```

5. Basic Example

```
<p>If compilation fails, terminal shows <samp>Error 404: File Not Found</samp>.</p>
```

6. Output

Renders "Error 404: File Not Found" in default browser monospace font.

7. Attributes

Attribute	Description	Example	Required/Optional
Global Attributes	Standard global attributes.	<code>class="console-output"</code>	Optional

8. Attribute Example

```
<samp class="terminal-text">Build Succeeded: 0 errors, 0 warnings</samp>
```

9. Real-World Example

- **Use Case:** Used in CLI tool documentations, server status monitoring dashboards, and dev tutorials.
- **Code:**

```
<p>Terminal response: <samp>PING google.com (142.250.190.46): 56 data bytes</samp></p>
```

10. Important Notes

Distinguishes program results (`<samp>`) from user typed commands (`<kbd>`) and source code (`<code>`).

11. Related Tags

Related to `<code>` , `<kbd>` , `<var>` .

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Not deprecated.

2.14 `<var>`

1. Tag Name

- **Tag Name:** `<var>`
- **Opening & Closing Form:** `<var>...</var>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<var>` element represents a variable name in mathematical expressions or programming context.

3. Purpose / Function

Styles variables in mathematical formulas or programming documentation in italicized form.

4. Syntax

```
<p>Equation: <var>x</var> = <var>y</var> + 2</p>
```

5. Basic Example

```
<p>In the equation <var>E</var> = <var>m</var><var>c</var><sup>2</sup>, <var>c</var> is the speed of light.</p>
```

6. Output

Renders "E", "m", and "c" as italicized mathematical variables: $E = m \cdot c^2$

7. Attributes

Attribute	Description	Example	Required/Optional
Global Attributes	Standard global attributes.	<code>class="math-var"</code>	Optional

8. Attribute Example

```
<p>Let <var class="variable">n</var> be the number of array elements.</p>
```

9. Real-World Example

- **Use Case:** Used in online engineering math portals (e.g. Wolfram Alpha, Khan Academy) and algorithm analysis articles.
- **Code:**

```
<p>Time complexity is  $O(n \log n)$ .</p>
```

10. Important Notes

Provides clear semantic context to math screen readers compared to plain italic `<i>`.

11. Related Tags

Related to `<code>`, `<sub>`, `<sup>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Not deprecated.

2.15 `<b>`

1. Tag Name

- **Tag Name:** `<b>`
- **Opening & Closing Form:** `<b>...</b>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<b>` element draws attention to text for utilitarian purposes without implying any extra importance, urgency, or emphasis.

3. Purpose / Function

Used to offset text stylistically (e.g. keywords in abstract, product names, lead sentence terms) in bold font.

4. Syntax

```
<b>Bold Text</b>
```

5. Basic Example

```
<p>The two main components are <b>Frontend</b> and <b>Backend</b> development.</p>
```

6. Output

Renders "Frontend" and "Backend" in bold font weight.

7. Attributes

Attribute	Description	Example	Required/Optional
Global Attributes	Standard global attributes.	<code>class="fw-bold"</code>	Optional

8. Attribute Example

```
<b class="highlight-term">Important Term:</b> Abstract Data Type
```

9. Real-World Example

- **Use Case:** Used in product summaries, dictionary entries, and bold keyword list items.
- **Code:**

```
<p><b>Note:</b> Bring your hall ticket to the exam room.</p>
```

10. Important Notes

Do not use `<b>` when text carries strong importance or urgency; use `<strong>` instead for accessibility.

11. Related Tags

Related to `<strong>`, `<i>`, `<em>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Redefined in HTML5 for stylistic bolding without strong importance.

2.16 `<strong>`

1. Tag Name

- **Tag Name:** `<strong>`
- **Opening & Closing Form:** `<strong>...</strong>`

- **Tag Type:** Paired / Container Tag

2. Definition

The `<strong>` tag indicates that its contents have strong importance, seriousness, or urgency.

3. Purpose / Function

Renders text in bold font while semantically marking it as critical information for screen readers.

4. Syntax

```
<strong>Crucial Alert Text</strong>
```

5. Basic Example

```
<p><strong>Warning:</strong> Submitting after deadline results in 0 marks.</p>
```

6. Output

Renders "Warning:" in bold weight, announced with emphasized audio pitch by screen reader software.

7. Attributes

Attribute	Description	Example	Required/Optional
Global Attributes	Standard global attributes.	<code>class="text-danger"</code>	Optional

8. Attribute Example

```
<strong class="alert-text">Danger: High Voltage Area</strong>
```

9. Real-World Example

- **Use Case:** Used in security warnings, critical form instructions, and vital terms on banking sites.
- **Code:**

```
<p><strong>DO NOT</strong> share your OTP password with anyone.</p>
```

10. Important Notes

Use `<strong>` when text importance affects semantic meaning; use `<b>` for generic bolding without special importance.

11. Related Tags

Related to `<b>`, `<em>`, `<mark>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Not deprecated.

2.17 `<i>`

1. Tag Name

- **Tag Name:** `<i>`
- **Opening & Closing Form:** `<i>...</i>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<i>` element represents a range of text set off from normal prose in an alternate voice or mood (e.g. technical terms, foreign words, thoughts, ship names).

3. Purpose / Function

Styles text in italics for idiomatic expressions, foreign terms, or taxonomy terms.

4. Syntax

```
<i>Italicized Text</i>
```

5. Basic Example

```
<p>The Latin term <i>Et cetera</i> is commonly written as etc.</p>
```

6. Output

Renders "Et cetera" in italic font style.

7. Attributes

Attribute	Description	Example	Required/Optional
Global Attributes	Standard global attributes.	<code>class="font-italic"</code>	Optional

8. Attribute Example

```
<i class="latin-phrase">Status quo</i>
```

9. Real-World Example

- **Use Case:** Used for scientific names (*Homo sapiens*), foreign phrases, book titles, and icon fonts (e.g. FontAwesome `<i class="fa fa-user"></i>`).
- **Code:**

```
<p>The scientific name of lion is <i>Panthera leo</i>.</p>
```

10. Important Notes

For semantic stress emphasis, use `<em>`. Use `<i>` for technical terms or foreign phrases without extra stress emphasis.

11. Related Tags

Related to `<em>`, `<cite>`, `<var>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Redefined in HTML5 as idiomatic/alternate voice text.

2.18 `<em>`

1. Tag Name

- **Tag Name:** `<em>`
- **Opening & Closing Form:** `<em>...</em>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<em>` tag marks text that has stress emphasis. Nesting `<em>` increases the degree of stress.

3. Purpose / Function

Changes the vocal emphasis of a sentence when spoken aloud, altering semantic meaning.

4. Syntax

```
<em>Emphasized Text</em>
```

5. Basic Example

```
<p>You <em>must</em> complete the prerequisite course first.</p>
```

6. Output

Renders "must" in italic font. Screen readers pronounce "must" with verbal stress emphasis.

7. Attributes

Attribute	Description	Example	Required/Optional
Global Attributes	Standard global attributes.	<code>class="stress"</code>	Optional

8. Attribute Example

```
<p>I <em>love</em> web programming!</p>
```

9. Real-World Example

- **Use Case:** Used in speech transcripts, persuasive copy, legal disclaimers, and instructional material.
- **Code:**

```
<p>Do <em>not</em> unplug the USB drive while transferring files.</p>
```

10. Important Notes

Use `<em>` for verbal stress emphasis; use `<i>` for foreign phrases, taxonomy names, or technical terms.

11. Related Tags

Related to `<i>`, `<strong>`, `<mark>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Not deprecated.

2.19 `<mark>`

1. Tag Name

- **Tag Name:** `<mark>`
- **Opening & Closing Form:** `<mark>...</mark>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<mark>` tag represents text highlighted or marked for reference purposes due to its relevance in another context.

3. Purpose / Function

Visually highlights text with a default yellow background, representing search result matches or highlighted study notes.

4. Syntax

```
<mark>Highlighted Text</mark>
```

5. Basic Example

```
<p>Search results for "HTML": Learning <mark>HTML</mark> tags and attributes.</p>
```

6. Output

Renders "HTML" with a bright yellow background highlight like a physical highlighter marker.

7. Attributes

Attribute	Description	Example	Required/Optional
Global Attributes	Standard global attributes.	<code>style="background: cyan;"</code>	Optional

8. Attribute Example

```
<mark style="background-color: #fef08a;">Exam Keyword</mark>
```

9. Real-World Example

- **Use Case:** Used in search engines to highlight matching query terms, PDF text highlight tools, and study notes.
- **Code:**

```
<p>Query: "Data". Match found: Introduction to <mark>Data</mark> Structures.</p>
```

10. Important Notes

Do not use `<mark>` purely for visual syntax highlighting of code; use CSS spans for code highlighting.

11. Related Tags

Related to `<strong>` , `<em>` , `<small>` .

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Standard HTML5 element.

2.20 `<small>`

1. Tag Name

- **Tag Name:** `<small>`
- **Opening & Closing Form:** `<small>...</small>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<small>` tag renders side comments and small print such as copyright text, legal disclaimers, licensing terms, or sub-text.

3. Purpose / Function

Reduces text font size by one size smaller (e.g. from medium to small) while semantically denoting fine print.

4. Syntax

```
<small>Fine print text</small>
```

5. Basic Example

```
<p>Course Fee: $500 <small>(Taxes applicable as per government rules)</small></p>
```

6. Output

Renders tax notice in smaller sub-text font size.

7. Attributes

Attribute	Description	Example	Required/Optional
Global Attributes	Standard global attributes.	<code>class="text-muted"</code>	Optional

8. Attribute Example

```
<small class="copyright-text">© 2026 University Portal. All rights reserved.</small>
```

9. Real-World Example

- **Use Case:** Used in page footers for copyright notices, terms of service disclaimers, and pricing fine print.
- **Code:**

```
<footer>
  <small>&copy; 2026 College Name. Registered trademark.</small>
</footer>
```

10. Important Notes

In HTML5, `<small>` has semantic meaning representing "fine print" or legal caveats, not just arbitrary font shrink.

11. Related Tags

Related to `<footer>`, `<sub>`, `<sup>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Redefined in HTML5 for legal fine print.

2.21 `<sub>`

1. Tag Name

- **Tag Name:** `<sub>`
- **Opening & Closing Form:** `<sub>...</sub>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<sub>` tag defines subscript text. Subscript text appears half a character below the normal line and is rendered in a smaller font.

3. Purpose / Function

Used for chemical formulas (e.g., H₂O), mathematical matrix indices (x₁), and footnotes.

4. Syntax

```
Text<sub>subscript</sub>
```

5. Basic Example

```
<p>Chemical formula for water is H<sub>2</sub>O and Glucose is C<sub>6</sub>H<sub>12</sub>O<sub>
```

6. Output

Renders "2" below H, "6", "12", "6" below C, H, O as chemical subscript formulas.

7. Attributes

Attribute	Description	Example	Required/Optional
Global Attributes	Standard global attributes.	<code>class="chem-sub"</code>	Optional

8. Attribute Example

```
<p>Matrix element: A<sub>i,j</sub></p>
```

9. Real-World Example

- **Use Case:** Used in chemistry education sites, scientific research databases, and math formula rendering engines.
- **Code:**

```
<p>Sulfuric acid formula: H<sub>2</sub>SO<sub>4</sub></p>
```

10. Important Notes

Ensure subscript font sizes do not break baseline line-height spacing in paragraphs.

11. Related Tags

Related to `<sup>`, `<var>`, `<code>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Not deprecated.

2.22 `<sup>`

1. Tag Name

- **Tag Name:** `<sup>`
- **Opening & Closing Form:** `<sup>...</sup>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<sup>` tag defines superscript text. Superscript text appears half a character above the normal line and is rendered in a smaller font.

3. Purpose / Function

Used for mathematical exponents (e.g., x^2), ordinal numbers (1st, 2nd), and citation references [1].

4. Syntax

```
Text<sup>superscript</sup>
```

5. Basic Example

```
<p>Pythagoras Theorem: a<sup>2</sup> + b<sup>2</sup> = c<sup>2</sup></p>
```

6. Output

Renders "2" as raised exponent superscripts: $a^2 + b^2 = c^2$

7. Attributes

Attribute	Description	Example	Required/Optional
Global Attributes	Standard global attributes.	<code>class="footnote-ref"</code>	Optional

8. Attribute Example

```
<p>Today is August 28<sup>th</sup>, 2026.</p>
```

9. Real-World Example

- **Use Case:** Used in Wikipedia reference citation tags ([1]), math web portals, and date formatting.
- **Code:**

```
<p>Einstein equation:  $E = mc^2$ </p>  
<p>See reference<sup>[1]</sup> for details.</p>
```

10. Important Notes

Crucial for proper mathematical and citation notation standards.

11. Related Tags

Related to `<sub>`, `<var>`, `<cite>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Not deprecated.

2.23 `<del>`

1. Tag Name

- **Tag Name:** `<del>`
- **Opening & Closing Form:** `<del>...</del>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<del>` tag represents text that has been deleted or removed from a document.

3. Purpose / Function

Displays text with a strikethrough line across it, signaling track changes or price discounts.

4. Syntax

```
<del cite="URL" datetime="YYYY-MM-DD">Deleted Text</del>
```

5. Basic Example

```
<p>Special Price: <del>$100</del> $75!</p>
```

6. Output

Renders "\$100" crossed out with a horizontal strikethrough line.

7. Attributes

Attribute	Description	Example	Required/Optional
<code>cite</code>	URL documenting the reason for deletion.	<code>cite="rev.html"</code>	Optional
<code>datetime</code>	Timestamp of when deletion occurred.	<code>datetime="2026-08-28T10:00"</code>	Optional

8. Attribute Example

```
<p>Exam Date: <del datetime="2026-08-01">August 10</del> <ins datetime="2026-08-01">August 15</ins></p>
```

9. Real-World Example

- **Use Case:** Used on E-commerce discount product cards, document versioning tools (GitHub diffs), and wiki edit logs.
- **Code:**

```
<p>Original Fee: <del>$500</del> <ins>$350</ins> (Discounted)</p>
```

10. Important Notes

Pair with `<ins>` tag for documenting document edits and versioning history.

11. Related Tags

Related to `<ins>`, `<s>`, `<mark>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Not deprecated.

2.24 `<ins>`

1. Tag Name

- **Tag Name:** `<ins>`
- **Opening & Closing Form:** `<ins>...</ins>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<ins>` tag represents a range of text that has been inserted into a document.

3. Purpose / Function

Displays inserted text with an underline, representing additions in document editing or revision history.

4. Syntax

```
<ins cite="URL" datetime="YYYY-MM-DD">Inserted Text</ins>
```

5. Basic Example

```
<p>Project submission deadline extended to <ins>Friday 5 PM</ins>.</p>
```

6. Output

Renders "Friday 5 PM" with a visual underline indicating new inserted content.

7. Attributes

Attribute	Description	Example	Required/Optional
<code>cite</code>	Specifies URL explaining edit reason.	<code>cite="edit-reason.html"</code>	Optional
<code>datetime</code>	Specifies date/time of insertion.	<code>datetime="2026-08-28"</code>	Optional

8. Attribute Example

```
<ins cite="notice.pdf" datetime="2026-08-28">New Syllabus updated</ins>
```

9. Real-World Example

- **Use Case:** Used in legal contract diff displays, GitHub code change reviews, and collaborative document editors.
- **Code:**

```
<p>The meeting is rescheduled from <del>2 PM</del> to <ins>4 PM</ins>.</p>
```

10. Important Notes

Always combine `<del>` and `<ins>` for semantic diff representation.

11. Related Tags

Related to `<del>`, `<u>`, `<s>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Not deprecated.

2.25 `<s>`

1. Tag Name

- **Tag Name:** `<s>`
- **Opening & Closing Form:** `<s>...</s>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<s>` tag renders text with a strikethrough line, representing content that is no longer accurate or relevant.

3. Purpose / Function

Used for out-of-date information or out-of-stock items where visual strikethrough is needed without document edit semantics.

4. Syntax

```
<s>Outdated text</s>
```

5. Basic Example

```
<p>Item Status: <s>Sold Out</s> (Back in Stock!)</p>
```

6. Output

Renders "Sold Out" crossed through with a line.

7. Attributes

Attribute	Description	Example	Required/Optional
Global Attributes	Standard global attributes.	<code>class="text-muted"</code>	Optional

8. Attribute Example

```
<s class="strikethrough">Old price $50</s>
```

9. Real-World Example

- **Use Case:** Used in shopping carts for strike-through original prices and expired promo codes.
- **Code:**

```
<p>Offer: <s>Free Shipping</s> (Standard rates apply)</p>
```

10. Important Notes

Use `<del>` when indicating actual historical document revisions; use `<s>` for general non-relevant text display.

11. Related Tags

Related to `<del>`, `<ins>`, `<u>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Redefined in HTML5.

2.26 `<u>`

1. Tag Name

- **Tag Name:** `<u>`
- **Opening & Closing Form:** `<u>...</u>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<u>` tag renders text with an unarticulated line (underline) to represent non-textual annotation such as spelling errors or proper names in Chinese.

3. Purpose / Function

Renders text with an underline without implying hyperlink functionality.

4. Syntax

```
<u>Annotated text</u>
```

5. Basic Example

```
<p>Please check for <u style="text-decoration-color: red;">speling</u> errors in your code.</p>
```

6. Output

Renders "speling" underlined (often styled with red squiggly line for spell check UI).

7. Attributes

Attribute	Description	Example	Required/Optional
Global Attributes	Standard global attributes.	<code>class="spelling-error"</code>	Optional

8. Attribute Example

```
<u class="wavy-underline">Unsure term</u>
```

9. Real-World Example

- **Use Case:** Used in word processor interfaces (Google Docs red spell check line) and language annotations.
- **Code:**

```
<p>Misspelled word detected: <u class="error-mark">teh</u> (Replace with "the")</p>
```

10. Important Notes

Avoid using `<u>` where users might confuse underlined text with a clickable hyperlink `<a>`. Style with CSS instead.

11. Related Tags

Related to `<a>`, `<ins>`, `<mark>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Redefined in HTML5 to avoid confusion with hyperlinks.

2.27 `<bdi>`

1. Tag Name

- **Tag Name:** `<bdi>`
- **Opening & Closing Form:** `<bdi>...</bdi>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<bdi>` (Bi-Directional Isolation) tag isolates a span of text that might be formatted in a different direction (e.g. Arabic, Hebrew) from surrounding text.

3. Purpose / Function

Prevents bidirectional text rendering bugs when embedding dynamic user-generated content in mixed language sites.

4. Syntax

```
<bdi>User generated text</bdi>
```

5. Basic Example

```
<ul>  
  <li>User <bdi>إبراهيم</bdi>: 50 points</li>  
  <li>User <bdi>John</bdi>: 45 points</li>  
</ul>
```

6. Output

Renders username "إبراهيم" correctly in right-to-left orientation without distorting surrounding colon and point text direction.

7. Attributes

Attribute	Description	Example	Required/Optional
Global Attributes	Standard global attributes.	<code>dir="auto"</code>	Optional

8. Attribute Example

```
<bdi class="username">Dynamic Name</bdi>
```

9. Real-World Example

- **Use Case:** Used on international forum platforms, Twitter/X usernames, and multilingual eCommerce comments.
- **Code:**

```
<p>Top Commenter: <bdi>سارة</bdi> liked your post.</p>
```

10. Important Notes

Essential for internationalization (i18n) when user-supplied names or titles are of unknown text direction.

11. Related Tags

Related to `<bdo>` and global `dir` attribute.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Standard HTML5 element.

2.28 `<bdo>`

1. Tag Name

- **Tag Name:** `<bdo>`
- **Opening & Closing Form:** `<bdo dir="ltr|rtl">...</bdo>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<bdo>` (Bi-Directional Override) element explicitly overrides the current text directionality.

3. Purpose / Function

Forces text to render strictly Left-To-Right (`ltr`) or Right-To-Left (`rtl`) regardless of Unicode properties.

4. Syntax

```
<bdo dir="ltr|rtl">Text content</bdo>
```

5. Basic Example

```
<p><bdo dir="rtl">This text will be written from right to left.</bdo></p>
```

6. Output

Renders text reversed character by character from right to left display.

7. Attributes

Attribute	Description	Example	Required/Optional
<code>dir</code>	Specifies mandatory text direction (<code>ltr</code> or <code>rtl</code>).	<code>dir="rtl"</code>	Required

8. Attribute Example

```
<bdo dir="rtl">123456789</bdo>
```

9. Real-World Example

- **Use Case:** Used in specialized multi-lingual typography rendering, RTL layout adjustments, and text games.
- **Code:**

```
<p>Reversed string display: <bdo dir="rtl">HTML5 Assignment</bdo></p>
```

10. Important Notes

The `dir` attribute is mandatory on `<bdo>`. Omitting `dir` invalidates the element.

11. Related Tags

Related to `<bdi>` and `lang` attribute.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Not deprecated.

2.29 `<wbr>`

1. Tag Name

- **Tag Name:** `<wbr>`
- **Opening & Closing Form:** `<wbr>` (No closing tag)
- **Tag Type:** Empty / Void Tag

2. Definition

The `<wbr>` (Word Break Opportunity) element specifies a position within text where the browser may optionally break a line if needed by wrapping rules.

3. Purpose / Function

Prevents awkward layout overflow for extremely long URL strings or unhyphenated compound words on mobile screens.

4. Syntax

```
VeryLongWord<wbr>SubPart
```

5. Basic Example

```
<p>Visit https://www.university<wbr>.ac.in/departments<wbr>/computer-science<wbr>/syllabus</p>
```

6. Output

Renders the long URL cleanly, breaking at `<wbr>` locations on small mobile viewports without forcing horizontal scrollbars.

7. Attributes

Attribute	Description	Example	Required/Optional
Global Attributes	Standard global attributes.	<code>class="break-opp"</code>	Optional

8. Attribute Example

```
<span>Supercalifragilistic<wbr>expialidocious</span>
```

9. Real-World Example

- **Use Case:** Used in mobile responsive web designs displaying long API endpoints, file paths, or DNA sequences.
- **Code:**

```
<p>File: C:/Users/Student/Desktop/Projects/WebTech/<wbr>Assignment_Final_Version_2026.html</p>
```

10. Important Notes

Unlike `­` (soft hyphen), `<wbr>` does not insert a hyphen character (-) when the line wraps.

11. Related Tags

Related to `<br>` and CSS `word-break` property.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Standard HTML5 element.

SECTION C: LINK AND NAVIGATION TAGS

3.1 `<a>`

1. Tag Name

- **Tag Name:** `<a>`
- **Opening & Closing Form:** `<a>...</a>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<a>` (Anchor) element creates a hyperlink to web pages, files, email addresses, locations within the same page, or any URL target.

3. Purpose / Function

Enables hyperlinking and navigation between web resources—the core foundational feature of the World Wide Web.

4. Syntax

```
<a href="URL" target="_blank|_self" rel="noopener">Anchor Text</a>
```

5. Basic Example

```
<a href="https://www.google.com" target="_blank" rel="noopener">Visit Google Search</a>
```

6. Output

Renders underlined blue text "Visit Google Search" which opens Google in a new browser tab when clicked.

7. Attributes

Attribute	Description	Example	Required/Optional
<code>href</code>	Specifies the destination URL or target anchor.	<code>href="https://univ.edu"</code>	Recommended
<code>target</code>	Browsing context to open link (<code>_blank</code> , <code>_self</code> , <code>_parent</code> , <code>_top</code>).	<code>target="_blank"</code>	Optional
<code>rel</code>	Specifies relationship between current doc and target (<code>noopener</code> , <code>noreferrer</code> , <code>nofollow</code>).	<code>rel="noopener"</code>	Recommended for <code>_blank</code>
<code>download</code>	Prompts browser to download target file rather than navigating to it.	<code>download="assignment.pdf"</code>	Optional
<code>hreflang</code>	Specifies language of linked document.	<code>hreflang="en"</code>	Optional
<code>type</code>	Specifies MIME type of target resource.	<code>type="application/pdf"</code>	Optional

8. Attribute Example

```
<a href="notes.pdf" download="HTML5_Notes.pdf" type="application/pdf">Download Study Notes (PDF)
```

9. Real-World Example

- **Use Case:** Used everywhere across all websites for site navigation menus, CTA buttons, internal page anchors, email links (`mailto:`), and tel links (`tel:`).
- **Code:**

```
<!-- Real world multi-attribute Anchor tag -->
<a href="mailto:support@college.edu?subject=Admission%20Query"
  class="btn btn-primary"
  title="Send Email to Support">
  Contact Support
</a>
```

10. Important Notes

Always add `rel="noopener noreferrer"` when using `target="_blank"` to prevent window.opener security phishing vulnerabilities. Ensure anchor text is descriptive for screen readers (avoid "click here").

11. Related Tags

Related to `<nav>` , `<link>` , `<button>` .

12. Deprecated/Obsolete Status

Standard HTML5 Tag. `name` attribute on `<a>` is obsolete in HTML5; use `id` attribute instead for page fragment anchors.

3.2 <nav>

1. Tag Name

- **Tag Name:** <nav>
- **Opening & Closing Form:** <nav>...</nav>
- **Tag Type:** Paired / Container Tag

2. Definition

The <nav> element represents a section of a page whose purpose is to provide navigation links, either within the current document or to other documents.

3. Purpose / Function

Semantically wraps main website navigation bars, sidebars, footers navigation, and pagination controls.

4. Syntax

```
<nav>
  <ul>
    <li><a href="index.html">Home</a></li>
    <li><a href="about.html">About</a></li>
  </ul>
</nav>
```

5. Basic Example

```
<nav aria-label="Main Menu">
  <a href="#home">Home</a> |
  <a href="#courses">Courses</a> |
  <a href="#contact">Contact</a>
</nav>
```

6. Output

Renders a navigation link group separated by pipe symbols.

7. Attributes

Attribute	Description	Example	Required/Optional
aria-label	Describes the navigation block for screen reader users.	aria-label="Primary Navigation"	Recommended
Global Attributes	Supports id , class , style .	class="navbar"	Optional

8. Attribute Example

```
<nav class="navbar navbar-expand-lg bg-light" aria-label="Top Menu">
  <!-- Navigation content -->
</nav>
```

9. Real-World Example

- **Use Case:** Used in header navigation bars of Amazon, Wikipedia, YouTube, and college portals.
- **Code:**

```
<header>
  <nav>
    <ul class="nav-menu">
      <li><a href="/dashboard">Dashboard</a></li>
      <li><a href="/courses">My Courses</a></li>
      <li><a href="/profile">Profile</a></li>
    </ul>
  </nav>
</header>
```

10. Important Notes

Not all links on a page should be wrapped in `<nav>`; it is reserved for major navigational blocks.

11. Related Tags

Related to `<a>`, `<header>`, `<ul>`, `<li>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Standard HTML5 semantic element.

SECTION D: IMAGES AND MULTIMEDIA TAGS

4.1 `<img>`

1. Tag Name

- **Tag Name:** `<img>`
- **Opening & Closing Form:** `<img ... >` (No closing tag)
- **Tag Type:** Empty / Void Tag

2. Definition

The `<img>` tag embeds an image into an HTML document. It creates a holding space for the referenced image file specified by the `src` attribute.

3. Purpose / Function

Displays graphical assets, logos, photographs, diagrams, and icons on web pages.

4. Syntax

```

```

5. Basic Example

```

```

6. Output

Renders the specified image on screen with width 600px and height 400px. If image fails to load, displays alternative text "University Belagavi Main Campus Building".

7. Attributes

Attribute	Description	Example	Required/Optional
<code>src</code>	Specifies path/URL of image file.	<code>src="logo.png"</code>	Required
<code>alt</code>	Alternative text description for screen readers and failed loads.	<code>alt="Company Logo"</code>	Required (a11y)
<code>width</code>	Specifies intrinsic width of image in pixels.	<code>width="300"</code>	Optional
<code>height</code>	Specifies intrinsic height of image in pixels.	<code>height="200"</code>	Optional
<code>loading</code>	Defines loading strategy (<code>lazy</code> or <code>eager</code>).	<code>loading="lazy"</code>	Recommended
<code>srcset</code>	Defines list of image sources for responsive screens.	<code>srcset="small.jpg 500w, large.jpg 1000w"</code>	Optional
<code>sizes</code>	Defines screen layout sizes for responsive images.	<code>sizes="(max-width: 600px) 100vw, 50vw"</code>	Optional
<code>decoding</code>	Hints image decoding (<code>async</code> , <code>sync</code> , <code>auto</code>).	<code>decoding="async"</code>	Optional

8. Attribute Example

```

```

10. Important Notes

The `alt` attribute is mandatory for accessibility compliance (WCAG). Always specify `width` and `height` to prevent Layout Shifts (CLS - Cumulative Layout Shift).

11. Related Tags

Related to `<picture>` , `<figure>` , `<figcaption>` , `<canvas>` .

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Attributes `align` , `border` , `hspace` , `vspace` are obsolete in HTML5.

4.2 <picture>

1. Tag Name

- **Tag Name:** <picture>
- **Opening & Closing Form:** <picture>...</picture>
- **Tag Type:** Paired / Container Tag

2. Definition

The <picture> tag contains zero or more <source> elements and one element to offer alternative versions of an image for different display devices/screen sizes.

3. Purpose / Function

Implements responsive art direction, serving different image formats (AVIF, WebP, JPG) or dimensions based on media query screen widths.

4. Syntax

```
<picture>
  <source media="(min-width: 800px)" srcset="large.jpg">
  <source media="(min-width: 450px)" srcset="medium.jpg">
  
</picture>
```

5. Basic Example

```
<picture>
  <source srcset="banner-dark.webp" media="(prefers-color-scheme: dark)">
  
</picture>
```

6. Output

Automatically displays dark theme banner when system dark mode is active; otherwise displays light mode banner.

7. Attributes

Attribute	Description	Example	Required/Optional
Global Attributes	Standard global attributes.	class="responsive-pic"	Optional

8. Attribute Example

```
<picture class="hero-image">
  <source srcset="hero.avif" type="image/avif">
  <source srcset="hero.webp" type="image/webp">
  
</picture>
```

9. Real-World Example

- **Use Case:** Used on modern news sites, media portals (Netflix, Apple.com) for crisp performance across mobile retina and desktop screens.
- **Code:**

```
<picture>
  <source media="(min-width: 1024px)" srcset="desktop-hero.jpg">
  <source media="(min-width: 640px)" srcset="tablet-hero.jpg">
  
</picture>
```

10. Important Notes

The fallback `<img>` element MUST be the last child element inside `<picture>`. If omitted, no image will be displayed.

11. Related Tags

Related to `<img>`, `<source>`, `<figure>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Standard HTML5 responsive element.

4.3 `<source>`

1. Tag Name

- **Tag Name:** `<source>`
- **Opening & Closing Form:** `<source ... >` (No closing tag)
- **Tag Type:** Empty / Void Tag

2. Definition

The `<source>` tag specifies multiple media resources for media elements like `<picture>`, `<audio>`, and `<video>`.

3. Purpose / Function

Provides alternative audio, video, or image format streams for browser codec fallback compatibility.

4. Syntax

```
<source src="file.mp4" type="video/mp4" media="(min-width: 600px)">
```

5. Basic Example

```
<video controls>
  <source src="lecture.webm" type="video/webm">
  <source src="lecture.mp4" type="video/mp4">
</video>
```

6. Output

Browser selects and plays WebM video format if supported, falling back to MP4 format seamlessly.

7. Attributes

Attribute	Description	Example	Required/Optional
src	Path to audio or video resource file.	src="audio.mp3"	Required for audio/video
srcset	Image source list for <picture> .	srcset="image.avif"	Required for picture
type	MIME type of media resource.	type="video/mp4"	Strongly Recommended
media	Media query condition for resource selection.	media="(min-width: 768px)"	Optional

8. Attribute Example

```
<source srcset="high-res.webp" type="image/webp" media="(min-width: 1200px)">
```

9. Real-World Example

- **Use Case:** Used in video platforms (YouTube embeds), responsive hero pictures, and podcasts.
- **Code:**

```
<audio controls>  
  <source src="podcast.opus" type="audio/ogg; codecs=opus">  
  <source src="podcast.mp3" type="audio/mpeg">  
</audio>
```

10. Important Notes

<source> tags are evaluated sequentially by browser engines from top to bottom; the first supported source is loaded.

11. Related Tags

Nested inside <picture> , <video> , and <audio> .

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Not deprecated.

4.4 <figure>

1. Tag Name

- **Tag Name:** <figure>
- **Opening & Closing Form:** <figure>...</figure>
- **Tag Type:** Paired / Container Tag

2. Definition

The <figure> tag represents self-contained content, frequently with a caption (<figcaption>), referenced as a single unit.

3. Purpose / Function

Encapsulates images, diagrams, photos, code snippets, or quotes alongside an optional caption.

4. Syntax

```
<figure>
  <!-- Media content like img, code, video -->
  <figcaption>Caption text</figcaption>
</figure>
```

5. Basic Example

```
<figure>
  
  <figcaption>Figure 1.1: Binary Search Tree Representation</figcaption>
</figure>
```

6. Output

Displays binary tree diagram image with caption text aligned beneath it as a unified semantic block.

7. Attributes

Attribute	Description	Example	Required/Optional
Global Attributes	Standard global attributes.	<code>class="diagram-box"</code>	Optional

8. Attribute Example

```
<figure class="figure border p-2 rounded">
  
  <figcaption class="figure-caption">Quarterly Growth Chart</figcaption>
</figure>
```

9. Real-World Example

- **Use Case:** Used in textbook publications, online academic journals, engineering blogs, and newspaper articles.
- **Code:**

```
<figure>
  <pre><code>int x = 10;</code></pre>
  <figcaption>Listing 1: Variable Declaration in C++</figcaption>
</figure>
```

10. Important Notes

Can contain multiple images or elements, but usually contains only one `<figcaption>` as first or last child.

11. Related Tags

Related to `<figcaption>`, `<img>`, `<blockquote>`, `<pre>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Standard HTML5 element.

4.5 <figcaption>

1. Tag Name

- **Tag Name:** <figcaption>
- **Opening & Closing Form:** <figcaption>...</figcaption>
- **Tag Type:** Paired / Container Tag

2. Definition

The <figcaption> tag represents a caption or legend describing the rest of the contents of its parent <figure> element.

3. Purpose / Function

Provides title labels, citations, or descriptive captions to diagrams, code listings, or images.

4. Syntax

```
<figcaption>Descriptive Caption Text</figcaption>
```

5. Basic Example

```
<figcaption>Fig. 3.2 – Architecture of von Neumann Computer Model</figcaption>
```

6. Output

Renders caption text underneath or above figure content in smaller/italic legend styling.

7. Attributes

Attribute	Description	Example	Required/Optional
Global Attributes	Standard global attributes.	class="caption-text"	Optional

8. Attribute Example

```
<figcaption class="text-center text-muted">Figure 1: Campus Infrastructure Map</figcaption>
```

9. Real-World Example

- **Use Case:** Used on Wikipedia articles, research papers, image galleries, and news photography.
- **Code:**

```
<figure>  
    
  <figcaption>Figure 4: Molecular structure of Deoxyribonucleic acid (DNA)</figcaption>  
</figure>
```

10. Important Notes

Must be placed as either the FIRST child or LAST child inside a `<figure>` tag.

11. Related Tags

Parent tag must be `<figure>`. Related to `<caption>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Not deprecated.

4.6 `<audio>`

1. Tag Name

- **Tag Name:** `<audio>`
- **Opening & Closing Form:** `<audio>...</audio>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<audio>` element is used to embed sound content in documents, such as music, podcasts, or voice recordings.

3. Purpose / Function

Plays audio files directly in web browsers natively without requiring external plugin software (Flash).

4. Syntax

```
<audio src="audio.mp3" controls>
  Your browser does not support audio.
</audio>
```

5. Basic Example

```
<audio controls preload="metadata">
  <source src="lecture-audio.mp3" type="audio/mpeg">
  <source src="lecture-audio.ogg" type="audio/ogg">
  Your browser does not support the audio element.
</audio>
```

6. Output

Displays native browser media player controls (Play/Pause button, timeline scrubber, volume slider, duration display).

7. Attributes

Attribute	Description	Example	Required/Optional
<code>src</code>	Specifies path to audio file.	<code>src="song.mp3"</code>	Optional if source nested
<code>controls</code>	Displays browser default playback UI controls.	<code>controls</code>	Recommended
<code>autoplay</code>	Plays audio automatically on page load (Muted policy applies).	<code>autoplay</code>	Optional
<code>loop</code>	Automatically replays audio when finished.	<code>loop</code>	Optional
<code>muted</code>	Mutes audio output by default.	<code>muted</code>	Optional
<code>preload</code>	Hints preloading strategy (<code>none</code> , <code>metadata</code> , <code>auto</code>).	<code>preload="none"</code>	Optional

8. Attribute Example

```
<audio controls loop muted preload="auto">
  <source src="background.mp3" type="audio/mp3">
</audio>
```

9. Real-World Example

- **Use Case:** Used on Spotify Web Player, SoundCloud, podcast websites, and online language learning apps (Duolingo).
- **Code:**

```
<div class="audio-player">
  <h3>Pronunciation Guide:</h3>
  <audio controls>
    <source src="word-pronunciation.mp3" type="audio/mpeg">
  </audio>
</div>
```

10. Important Notes

Modern browsers block `autoplay` with sound enabled unless user has previously interacted with the web page.

11. Related Tags

Related to `<video>` , `<source>` , `<track>` .

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Replaced legacy `<embed>` and Flash audio plugins.

4.7 `<video>`

1. Tag Name

- **Tag Name:** `<video>`

- **Opening & Closing Form:** `<video>...</video>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<video>` element embeds a media player for video playback (e.g. MP4, WebM) natively inside HTML documents.

3. Purpose / Function

Renders video streams, online video lectures, movie trailers, and background video animations.

4. Syntax

```
<video src="video.mp4" controls width="640" height="360"></video>
```

5. Basic Example

```
<video width="800" height="450" controls poster="thumbnail.jpg">  
  <source src="lab-demo.mp4" type="video/mp4">  
  <source src="lab-demo.webm" type="video/webm">  
  Your browser does not support HTML5 video.  
</video>
```

6. Output

Displays an 800x450 video player box showing thumbnail image, play button, volume slider, full-screen toggle, and scrub timeline.

7. Attributes

Attribute	Description	Example	Required/Optional
<code>src</code>	Specifies URL of video file.	<code>src="movie.mp4"</code>	Optional if source nested
<code>controls</code>	Shows playback controls (play, pause, volume, fullscreen).	<code>controls</code>	Recommended
<code>poster</code>	Specifies preview thumbnail image displayed before play.	<code>poster="thumb.jpg"</code>	Optional
<code>width</code>	Width of video player box in pixels.	<code>width="640"</code>	Recommended
<code>height</code>	Height of video player box in pixels.	<code>height="360"</code>	Recommended
<code>autoplay</code>	Plays video automatically upon load.	<code>autoplay</code>	Optional
<code>loop</code>	Loops video playback continuously.	<code>loop</code>	Optional
<code>muted</code>	Silences audio track of video.	<code>muted</code>	Recommended for bg videos
<code>playsinline</code>	Plays inline on mobile browsers instead of forced fullscreen.	<code>playsinline</code>	Recommended for mobile

8. Attribute Example

```
<video width="100%" height="auto" autoplay loop muted playsinline poster="bg-poster.jpg">
  <source src="hero-loop.mp4" type="video/mp4">
</video>
```

9. Real-World Example

- **Use Case:** Used in video streaming platforms (YouTube, Netflix, Vimeo), course portals (Coursera, Udemy), and landing page hero backgrounds.
- **Code:**

```
<section class="course-video">
  <h2>Lecture 1: Introduction to Data Structures</h2>
  <video controls width="100%" poster="lecture1-cover.png">
    <source src="https://cdn.univ.edu/lectures/lec1.mp4" type="video/mp4">
    <track src="subtitles_en.vtt" kind="subtitles" srclang="en" label="English">
  </video>
</section>
```

10. Important Notes

To enable `autoplay` reliably across mobile and desktop browsers, you MUST also add the `muted` attribute.

11. Related Tags

Related to `<audio>`, `<source>`, `<track>`, `<iframe>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Replaced obsolete `<object>` and Flash video players.

4.8 `<track>`

1. Tag Name

- **Tag Name:** `<track>`
- **Opening & Closing Form:** `<track ... >` (No closing tag)
- **Tag Type:** Empty / Void Tag

2. Definition

The `<track>` tag is used as a child of `<audio>` and `<video>` elements to specify timed text tracks (e.g. subtitles, captions, chapter headings).

3. Purpose / Function

Adds closed captions (CC), subtitles in multiple languages, or chapter definitions to video/audio players for accessibility.

4. Syntax

```
<track kind="subtitles|captions" src="file.vtt" srclang="en" label="English">
```

5. Basic Example

```
<video controls src="lecture.mp4">
  <track kind="subtitles" src="subtitles_en.vtt" srclang="en" label="English Subtitles" default>
  <track kind="subtitles" src="subtitles_hi.vtt" srclang="hi" label="Hindi Subtitles">
</video>
```

6. Output

Overlay closed caption subtitle text on top of playing video based on timing defined in WebVTT (`.vtt`) file.

7. Attributes

Attribute	Description	Example	Required/Optional
<code>src</code>	URL of WebVTT track file (.vtt).	<code>src="subtitles.vtt"</code>	Required
<code>kind</code>	Type of text track (<code>subtitles</code> , <code>captions</code> , <code>descriptions</code> , <code>chapters</code>).	<code>kind="captions"</code>	Required
<code>srclang</code>	Two-letter language code of track text.	<code>srclang="en"</code>	Required for subtitles
<code>label</code>	User-visible title of track in player CC selection menu.	<code>label="English (CC)"</code>	Recommended
<code>default</code>	Enables this track by default if user preferences match.	<code>default</code>	Optional

8. Attribute Example

```
<track kind="chapters" src="chapters.vtt" srclang="en" label="Chapter Markers">
```

9. Real-World Example

- **Use Case:** Used on YouTube, Netflix, Coursera, and TED Talks for international subtitles and deaf accessibility captions.
- **Code:**

```
<video controls poster="movie.jpg">
  <source src="movie.mp4" type="video/mp4">
  <track kind="captions" src="movie_en.vtt" srclang="en" label="English Captions" default>
</video>
```

10. Important Notes

Must point to valid WebVTT standard format file (`.vtt`). Child tag of `<video>` or `<audio>` .

11. Related Tags

Nested inside `<video>` and `<audio>` .

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Standard HTML5 accessibility element.

4.9 <iframe>

1. Tag Name

- **Tag Name:** <iframe>
- **Opening & Closing Form:** <iframe></iframe>
- **Tag Type:** Paired / Container Tag

2. Definition

The <iframe> (Inline Frame) tag embeds another HTML document into the current web page.

3. Purpose / Function

Embeds external third-party widgets like Google Maps, YouTube video embeds, payment gateways, or external page previews.

4. Syntax

```
<iframe src="URL" width="width" height="height" title="Description"></iframe>
```

5. Basic Example

```
<iframe src="https://maps.google.com/maps?q=belagavi&output=embed" width="600" height="450" styl
```

6. Output

Renders an interactive Google Map window embedded directly inside current webpage container.

7. Attributes

Attribute	Description	Example	Required/Optional
src	Address of document to embed.	src="page.html"	Required
width	Frame width in pixels or %.	width="560"	Optional
height	Frame height in pixels or %.	height="315"	Optional
title	Accessible text label for screen readers.	title="YouTube Player"	Mandatory for a11y
sandbox	Enables strict security restrictions on iframe content.	sandbox="allow-scripts"	Strongly Recommended
allow	Specifies feature policy capabilities (fullscreen , camera).	allow="fullscreen"	Optional
loading	Lazy load frame (lazy , eager).	loading="lazy"	Optional

8. Attribute Example

```
<iframe src="https://www.youtube-nocookie.com/embed/dQw4w9WgXcQ" width="560" height="315" title=
```

9. Real-World Example

- **Use Case:** Used for YouTube video embeds, Google Maps on contact pages, Stripe payment forms, and CodePen embeds.
- **Code:**

```
<div class="ratio ratio-16x9">  
  <iframe src="https://www.youtube.com/embed/dQw4w9WgXcQ" title="College Intro Video" allowfulls  
</div>
```

10. Important Notes

Always include a descriptive `title` attribute for screen readers. Use `sandbox` attribute when embedding untrusted third-party URLs to prevent malicious script executions.

11. Related Tags

Related to `<video>`, `<embed>`, `<object>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Attributes `frameborder`, `scrolling`, `marginwidth`, `marginheight` are obsolete in HTML5.

SECTION E: LIST TAGS

5.1

1. Tag Name

- **Tag Name:**
- **Opening & Closing Form:** ...
- **Tag Type:** Paired / Container Tag

2. Definition

The tag defines an unordered list of items. Items in are displayed with bullet point symbols by default.

3. Purpose / Function

Groups related items where order of sequence does not matter, such as feature lists, navigation links, or ingredient lists.

4. Syntax

```
<ul>
  <li>Item 1</li>
  <li>Item 2</li>
</ul>
```

5. Basic Example

```
<ul>
  <li>Data Structures</li>
  <li>Operating Systems</li>
  <li>Web Technologies</li>
</ul>
```

6. Output

Renders a bulleted list with solid circular bullet dots preceding each course item.

7. Attributes

Attribute	Description	Example	Required/Optional
type	Sets bullet style (disc , circle , square).	type="square"	Deprecated (Use CSS list-style-type)
Global Attributes	Standard global attributes.	class="feature-list"	Optional

8. Attribute Example

```
<ul style="list-style-type: square;" class="ms-4">
  <li>HTML5</li>
  <li>CSS3</li>
</ul>
```

9. Real-World Example

- **Use Case:** Used in navigation menus (`<nav><ul>`), footer link columns, feature bullet lists, and bulleted assignment topics.
- **Code:**

```
<nav>
  <ul class="nav-links">
    <li><a href="/home">Home</a></li>
    <li><a href="/about">About</a></li>
  </ul>
</nav>
```

10. Important Notes

Direct child elements of `<ul>` MUST only be `<li>` elements (or `<template>` / `<script>`). Never put `<p>` or `<div>` directly inside `<ul>` .

11. Related Tags

Related to `<ol>` , `<li>` , `<dl>` .

12. Deprecated/Obsolete Status

Standard HTML5 Tag. `type` attribute is deprecated in HTML5; use CSS `list-style-type` .

5.2 `<ol>`

1. Tag Name

- **Tag Name:** `<ol>`
- **Opening & Closing Form:** `<ol>...</ol>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<ol>` tag defines an ordered list of items. Items in `<ol>` are automatically numbered sequentially by default.

3. Purpose / Function

Presents sequential step-by-step instructions, rankings, recipes, or prioritized task lists.

4. Syntax

```
<ol type="1|A|a|I|i" start="number">
  <li>First step</li>
  <li>Second step</li>
</ol>
```

5. Basic Example

```
<ol>
  <li>Download VS Code Editor</li>
  <li>Install Live Server Extension</li>
  <li>Write index.html file</li>
</ol>
```

6. Output

Renders a numbered list: 1. Download VS Code... 2. Install Live Server... 3. Write index.html...

7. Attributes

Attribute	Description	Example	Required/Optional
type	Specifies numbering type (1 numbers, A uppercase letters, a lowercase letters, I uppercase Roman, i lowercase Roman).	type="A"	Optional
start	Specifies starting numerical value for list sequence.	start="5"	Optional
reversed	Reverses numbering sequence order (high to low).	reversed	Optional

8. Attribute Example

```
<ol type="I" start="1" class="syllabus-list">
  <li>Module I: Basics</li>
  <li>Module II: Advanced</li>
</ol>
```

9. Real-World Example

- **Use Case:** Used in recipe instruction steps, top 10 rankings, algorithmic step listings, and legal document clauses.
- **Code:**

```
<h3>Algorithm Steps:</h3>
<ol type="1">
  <li>Start program</li>
  <li>Read inputs <var>a</var> and <var>b</var></li>
  <li>Compute sum = <var>a</var> + <var>b</var></li>
  <li>Print sum and stop</li>
</ol>
```

10. Important Notes

Direct children of `<ol>` MUST only be `<li>` elements.

11. Related Tags

Related to `<ul>`, `<li>`, `<dl>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. `compact` attribute is obsolete.

5.3 `<li>`

1. Tag Name

- **Tag Name:** `<li>`
- **Opening & Closing Form:** `<li>...</li>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<li>` (List Item) tag is used to represent an individual item inside an ordered list (`<ol>`), unordered list (`<ul>`), or menu (`<menu>`).

3. Purpose / Function

Wraps content for each individual item entry within a parent list container.

4. Syntax

```
<li>List item content</li>
```

5. Basic Example

```
<ul>
  <li>C++ Programming</li>
  <li>Java Programming</li>
</ul>
```

6. Output

Renders an individual bulleted list item entry.

7. Attributes

Attribute	Description	Example	Required/Optional
<code>value</code>	Sets ordinal numerical value for item in an <code></code> .	<code>value="10"</code>	Optional (Only in <code></code>)
<code>type</code>	Sets marker style for this specific item.	<code>type="circle"</code>	Deprecated

8. Attribute Example

```
<ol>
  <li value="5">Step 5 (Resumed)</li>
  <li>Step 6</li>
</ol>
```

9. Real-World Example

- **Use Case:** Used in every dropdown menu option, navbar link list, step procedure, and tab panel item.
- **Code:**

```
<ul class="todo-list">
  <li class="completed">Finish HTML Assignment</li>
  <li class="pending">Study Operating Systems</li>
</ul>
```

10. Important Notes

Parent container MUST be `<ul>`, `<ol>`, or `<menu>`. Can contain nested `<ul>` or `<ol>` for multi-level hierarchical trees.

11. Related Tags

Nested inside `<ul>`, `<ol>`, `<menu>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. `type` attribute is deprecated.

5.4 `<dl>`

1. Tag Name

- **Tag Name:** `<dl>`
- **Opening & Closing Form:** `<dl>...</dl>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<dl>` tag defines a description list (formerly definition list), containing a set of terms (`<dt>`) and descriptions (`<dd>`).

3. Purpose / Function

Groups key-value pairs, glossary terms, metadata lists, or FAQ items.

4. Syntax

```
<dl>
  <dt>Term 1</dt>
  <dd>Description of Term 1</dd>
</dl>
```

5. Basic Example

```
<dl>
  <dt>HTML</dt>
  <dd>HyperText Markup Language</dd>
  <dt>CSS</dt>
  <dd>Cascading Style Sheets</dd>
</dl>
```

6. Output

Displays terms bold/left-aligned and definitions indented underneath each term.

7. Attributes

Attribute	Description	Example	Required/Optional
Global Attributes	Standard global attributes.	<code>class="glossary"</code>	Optional

8. Attribute Example

```
<dl class="row">
  <dt class="col-sm-3">Course Code</dt>
  <dd class="col-sm-9">18CS52</dd>
</dl>
```

9. Real-World Example

- **Use Case:** Used for glossaries, product specification key-value tables, metadata sidebars, and FAQs.
- **Code:**

```
<dl>
  <dt>CPU</dt>
  <dd>Central Processing Unit - Main execution hardware.</dd>
  <dt>RAM</dt>
  <dd>Random Access Memory - Volatile system memory.</dd>
</dl>
```

10. Important Notes

Contains alternating `<dt>` (term) and `<dd>` (description) children. In HTML5, `<div>` tags can wrap `<dt>` / `<dd>` pairs inside `<dl>` for styling.

11. Related Tags

Related to `<dt>`, `<dd>`, `<ul>`, `<ol>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Not deprecated.

5.5 `<dt>`

1. Tag Name

- **Tag Name:** `<dt>`

- **Opening & Closing Form:** `<dt>...</dt>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<dt>` tag specifies a term (or name) in a description list (`<d1>`).

3. Purpose / Function

Acts as the key/term header in key-value description lists.

4. Syntax

```
<dt>Term Name</dt>
```

5. Basic Example

```
<dt>Bandwidth</dt>
```

6. Output

Displays "Bandwidth" in bold font weight.

7. Attributes

Attribute	Description	Example	Required/Optional
Global Attributes	Standard global attributes.	<code>class="fw-bold"</code>	Optional

8. Attribute Example

```
<dt class="term-title">Algorithm</dt>
```

9. Real-World Example

- **Use Case:** Used in glossaries, FAQ questions, dictionary terms, and property names.
- **Code:**

```
<d1>  
  <dt>Recursion</dt>  
  <dd>A process in which a function calls itself directly or indirectly.</dd>  
</d1>
```

10. Important Notes

Must be placed inside a `<d1>` element (or inside a `<div>` inside a `<d1>`). Followed by one or more `<dd>` tags.

11. Related Tags

Parent tag is `<d1>` . Sibling tag is `<dd>` .

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Not deprecated.

5.6 <dd>

1. Tag Name

- **Tag Name:** <dd>
- **Opening & Closing Form:** <dd>...</dd>
- **Tag Type:** Paired / Container Tag

2. Definition

The <dd> tag provides the description, definition, or value for the preceding term (<dt>) in a description list (<dl>).

3. Purpose / Function

Provides explanation text for terms defined by <dt> .

4. Syntax

```
<dd>Description text</dd>
```

5. Basic Example

```
<dd>Data rate supported by a network connection.</dd>
```

6. Output

Displays description text indented below or next to the preceding term.

7. Attributes

Attribute	Description	Example	Required/Optional
Global Attributes	Standard global attributes.	<code>class="desc-text"</code>	Optional

8. Attribute Example

```
<dd class="ms-3 text-muted">A step-by-step procedure for solving a problem.</dd>
```

9. Real-World Example

- **Use Case:** Used in dictionary definitions, product spec sheets (e.g. Dimensions: 15x10 cm), and FAQ answers.
- **Code:**

```
<dl>  
  <dt>How do I register?</dt>  
  <dd>Click on the sign up button and submit your details.</dd>  
</dl>
```

10. Important Notes

Must be a child of <dl> (or wrapper <div> inside <dl>). One <dt> can have multiple <dd> descriptions.

11. Related Tags

Parent is `<d1>`. Paired with `<dt>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Not deprecated.

SECTION F: TABLE TAGS

6.1 `<table>`

1. Tag Name

- **Tag Name:** `<table>`
- **Opening & Closing Form:** `<table>...</table>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<table>` HTML element represents tabular data — information presented in a two-dimensional table comprised of rows and columns.

3. Purpose / Function

Displays structured matrix data such as timetables, grade cards, pricing plans, and financial reports.

4. Syntax

```
<table>
  <tr>
    <th>Header</th>
  </tr>
  <tr>
    <td>Data</td>
  </tr>
</table>
```

5. Basic Example

```
<table border="1">
  <tr>
    <th>USN</th>
    <th>Name</th>
  </tr>
  <tr>
    <td>1VU22CS001</td>
    <td>Aryansh</td>
  </tr>
</table>
```

6. Output

Renders a grid table with border showing USN and Name headers with data row below.

7. Attributes

Attribute	Description	Example	Required/Optional
<code>border</code>	Enables simple table grid border (<code>0</code> or <code>1</code>).	<code>border="1"</code>	Optional (CSS preferred)
<code>cellpadding</code>	Pixel space inside cells.	<code>cellpadding="8"</code>	Deprecated (Use CSS padding)
<code>cellspacing</code>	Pixel space between cells.	<code>cellspacing="0"</code>	Deprecated (Use CSS border-spacing)
<code>width</code>	Specifies table width.	<code>width="100%"</code>	Deprecated (Use CSS width)

8. Attribute Example

```
<table class="table table-bordered table-striped" id="student-table">
```

9. Real-World Example

- **Use Case:** Used for college exam timetables, student grade reports, flight schedules, stock market tables.
- **Code:**

```
<table class="table-styled">
  <caption>Semester 3 Marks</caption>
  <thead><tr><th>Subject</th><th>Marks</th></tr></thead>
  <tbody><tr><td>Web Tech</td><td>95</td></tr></tbody>
</table>
```

10. Important Notes

Tables should ONLY be used for tabular data, NEVER for overall web page layout design (use CSS Flexbox/Grid instead).

11. Related Tags

Related to `<caption>`, `<thead>`, `<tbody>`, `<tfoot>`, `<tr>`, `<th>`, `<td>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Presentation attributes (`cellpadding`, `cellspacing`, `width`, `bgcolor`) are obsolete.

6.2 `<caption>`

1. Tag Name

- **Tag Name:** `<caption>`
- **Opening & Closing Form:** `<caption>...</caption>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<caption>` element specifies the title or heading of a table.

3. Purpose / Function

Provides a clear descriptive title for a table for accessibility screen readers.

4. Syntax

```
<caption>Table Title Caption</caption>
```

5. Basic Example

```
<table>
  <caption>Table 1.1: Student Attendance Record - August 2026</caption>
  <!-- rows -->
</table>
```

6. Output

Renders title text centered directly above the top border of the table.

7. Attributes

Attribute	Description	Example	Required/Optional
<code>align</code>	Sets caption placement (<code>top</code> , <code>bottom</code>).	<code>align="bottom"</code>	Deprecated

8. Attribute Example

```
<caption style="caption-side: bottom;" class="text-muted">Source: VTU Examination Board</caption>
```

9. Real-World Example

- **Use Case:** Used in academic journals, scientific data tables, and public report tables.
- **Code:**

```
<table>
  <caption>Weekly Class Time Table - CSE Sec A</caption>
  <tr><th>Day</th><th>9 AM - 10 AM</th></tr>
</table>
```

10. Important Notes

Must be the VERY FIRST child element inside a `<table>` tag, appearing before `<thead>` or `<tr>` .

11. Related Tags

Parent element must be `<table>` . Related to `<figcaption>` .

12. Deprecated/Obsolete Status

Standard HTML5 Tag. `align` attribute is deprecated.

6.3 <thead>

1. Tag Name

- **Tag Name:** <thead>
- **Opening & Closing Form:** <thead>...</thead>
- **Tag Type:** Paired / Container Tag

2. Definition

The <thead> (Table Head) tag groups the header content in an HTML table.

3. Purpose / Function

Encapsulates heading rows (<tr><th>), allowing browsers/printers to repeat table headers across multi-page printouts.

4. Syntax

```
<thead>
  <tr>
    <th>Column 1</th>
    <th>Column 2</th>
  </tr>
</thead>
```

5. Basic Example

```
<table>
  <thead>
    <tr>
      <th>Subject Code</th>
      <th>Subject Title</th>
      <th>Credits</th>
    </tr>
  </thead>
</table>
```

6. Output

Groups header row cleanly at top of table.

7. Attributes

Attribute	Description	Example	Required/Optional
Global Attributes	Standard global attributes.	class="table-dark"	Optional

8. Attribute Example

```
<thead class="bg-primary text-white">
```

9. Real-World Example

- **Use Case:** Used in large enterprise data tables, invoice printouts, and pagination tables.
- **Code:**

```
<table class="data-table">
  <thead class="sticky-top">
    <tr><th>ID</th><th>Name</th><th>Role</th></tr>
  </thead>
</table>
```

10. Important Notes

Must be placed inside `<table>`, after `<caption>`, and before `<tbody>` or `<tfoot>`.

11. Related Tags

Nested in `<table>`. Related to `<tbody>`, `<tfoot>`, `<th>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Not deprecated.

6.4 `<tbody>`

1. Tag Name

- **Tag Name:** `<tbody>`
- **Opening & Closing Form:** `<tbody>...</tbody>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<tbody>` (Table Body) tag encapsulates the main data rows (`<tr><td>`) of an HTML table.

3. Purpose / Function

Separates data rows from header (`<thead>`) and summary footer (`<tfoot>`) sections.

4. Syntax

```
<tbody>
  <tr>
    <td>Data 1</td>
    <td>Data 2</td>
  </tr>
</tbody>
```

5. Basic Example

```
<table>
  <tbody>
    <tr><td>18CS51</td><td>Management</td><td>3</td></tr>
    <tr><td>18CS52</td><td>Web Tech</td><td>4</td></tr>
  </tbody>
</table>
```

6. Output

Groups content rows of table body.

7. Attributes

Attribute	Description	Example	Required/Optional
Global Attributes	Standard global attributes.	<code>class="table-group-divider"</code>	Optional

8. Attribute Example

```
<tbody class="divide-y divide-gray-200">
```

9. Real-World Example

- **Use Case:** Used in dynamic JavaScript data tables (DataTables, React tables) for sorting and filtering data rows.
- **Code:**

```
<table>
  <thead><tr><th>Item</th><th>Price</th></tr></thead>
  <tbody id="cart-items">
    <tr><td>Laptop</td><td>$1000</td></tr>
  </tbody>
</table>
```

10. Important Notes

Multiple `<tbody>` elements can exist in a single table to divide data into distinct row sections.

11. Related Tags

Nested in `<table>` . Related to `<thead>` , `<tfoot>` , `<tr>` .

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Not deprecated.

6.5 `<tfoot>`

1. Tag Name

- **Tag Name:** `<tfoot>`
- **Opening & Closing Form:** `<tfoot>...</tfoot>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<tfoot>` (Table Footer) tag groups footer/summary content in an HTML table (such as totals, averages, or page counts).

3. Purpose / Function

Renders table summary footers at the bottom of data grids.

4. Syntax

```
<tfoot>
  <tr>
    <td>Total</td>
    <td>100</td>
  </tr>
</tfoot>
```

5. Basic Example

```
<table>
  <tfoot>
    <tr>
      <td>Total Credits:</td>
      <td>24</td>
    </tr>
  </tfoot>
</table>
```

6. Output

Displays total credits row at bottom of table.

7. Attributes

Attribute	Description	Example	Required/Optional
Global Attributes	Standard global attributes.	<code>class="fw-bold"</code>	Optional

8. Attribute Example

```
<tfoot class="bg-light font-weight-bold">
```

9. Real-World Example

- **Use Case:** Used in invoice balance sheets, e-commerce cart totals, and financial summaries.
- **Code:**

```
<table>
  <tbody><tr><td>Subtotal</td><td>$500</td></tr></tbody>
  <tfoot><tr><th>Grand Total:</th><th>$500</th></tr></tfoot>
</table>
```

10. Important Notes

In HTML5, `<tfoot>` can be placed either before or after `<tbody>`, but browser renders it at the bottom.

11. Related Tags

Nested in `<table>`. Related to `<thead>`, `<tbody>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Not deprecated.

6.6 <tr>

1. Tag Name

- **Tag Name:** <tr>
- **Opening & Closing Form:** <tr>...</tr>
- **Tag Type:** Paired / Container Tag

2. Definition

The <tr> (Table Row) tag defines a horizontal row of cells in a table.

3. Purpose / Function

Acts as container for table header cells (<th>) or table data cells (<td>).

4. Syntax

```
<tr>
  <th>Header</th>
  <td>Data</td>
</tr>
```

5. Basic Example

```
<tr>
  <td>August 28</td>
  <td>HTML5 Exam</td>
  <td>Passed</td>
</tr>
```

6. Output

Renders a single horizontal row containing three table cells.

7. Attributes

Attribute	Description	Example	Required/Optional
Global Attributes	Standard global attributes.	class="table-row-hover"	Optional

8. Attribute Example

```
<tr class="bg-gray-100" id="row-1">
```

9. Real-World Example

- **Use Case:** Used in every grid table row on web applications.
- **Code:**

```
<tbody>
  <tr id="student-101">
    <td>101</td>
    <td>Rahul Kumar</td>
  </tr>
</tbody>
```

10. Important Notes

Must be placed inside `<thead>`, `<tbody>`, `<tfoot>`, or directly in `<table>`. Contains only `<th>` or `<td>`.

11. Related Tags

Nested in `<thead>`, `<tbody>`, `<tfoot>`. Contains `<th>`, `<td>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. `bgcolor`, `align`, `valign` attributes are obsolete.

6.7 `<th>`

1. Tag Name

- **Tag Name:** `<th>`
- **Opening & Closing Form:** `<th>...</th>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<th>` (Table Header) tag defines a header cell in a table. Text inside `<th>` is bold and centered by default.

3. Purpose / Function

Provides column or row header labels in tabular data for human readability and screen reader accessibility.

4. Syntax

```
<th scope="col|row" colspan="n" rowspan="n">Header Title</th>
```

5. Basic Example

```
<tr>
  <th scope="col">Student ID</th>
  <th scope="col">Name</th>
  <th scope="col">GPA</th>
</tr>
```

6. Output

Displays "Student ID", "Name", "GPA" in bold, centered text inside header cells.

7. Attributes

Attribute	Description	Example	Required/Optional
scope	Specifies whether header applies to <code>col</code> , <code>row</code> , <code>colgroup</code> , or <code>rowgroup</code> .	<code>scope="col"</code>	Recommended (a11y)
colspan	Number of columns this header cell spans.	<code>colspan="2"</code>	Optional
rowspan	Number of rows this header cell spans.	<code>rowspan="2"</code>	Optional
abbr	Abbreviation for long header text.	<code>abbr="Sub Code"</code>	Optional

8. Attribute Example

```
<th scope="col" colspan="2" class="text-center">Contact Details</th>
```

9. Real-World Example

- **Use Case:** Used across all tabular data headers on college portals, banking sites, and data sheets.
- **Code:**

```
<thead>
  <tr>
    <th scope="col">Roll No</th>
    <th scope="col">Student Name</th>
  </tr>
</thead>
```

10. Important Notes

Always include `scope="col"` or `scope="row"` on `<th>` elements to ensure screen readers correctly associate headers with data cells.

11. Related Tags

Nested in `<tr>` . Related to `<td>` , `<thead>` .

12. Deprecated/Obsolete Status

Standard HTML5 Tag. `align` , `width` , `height` , `bgcolor` obsolete.

6.8 <td>

1. Tag Name

- **Tag Name:** `<td>`
- **Opening & Closing Form:** `<td>...</td>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<td>` (Table Data) tag defines a standard data cell in an HTML table. Text in `<td>` is regular-weighted and left-aligned by default.

3. Purpose / Function

Contains actual data values within table grid rows.

4. Syntax

```
<td colspan="n" rowspan="n">Cell Data</td>
```

5. Basic Example

```
<tr>
  <td>1VU22CS045</td>
  <td>Priya Sharma</td>
  <td>9.4</td>
</tr>
```

6. Output

Renders three regular left-aligned data cells inside a table row.

7. Attributes

Attribute	Description	Example	Required/Optional
colspan	Specifies number of columns cell should span across.	colspan="3"	Optional
rowspan	Specifies number of rows cell should span across.	rowspan="2"	Optional
headers	Space-separated IDs of <th> elements providing header context.	headers="col-name"	Optional

8. Attribute Example

```
<td colspan="2" class="text-center font-bold">Total Marks: 450</td>
```

9. Real-World Example

- **Use Case:** Used to display cell data values in grade lists, pricing grid comparisons, and financial reports.
- **Code:**

```
<tr>
  <td>Web Tech Lab</td>
  <td rowspan="2">Exam Slot A</td>
</tr>
<tr>
  <td>OS Lab</td>
</tr>
```

10. Important Notes

Use colspan and rowspan for merging adjacent table cells in complex schedule tables.

11. Related Tags

Nested in <tr> . Related to <th> , <table> .

12. Deprecated/Obsolete Status

Standard HTML5 Tag. `width`, `height`, `align`, `valign`, `bgcolor` obsolete.

6.9 `<colgroup>`

1. Tag Name

- **Tag Name:** `<colgroup>`
- **Opening & Closing Form:** `<colgroup>...</colgroup>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<colgroup>` tag specifies a group of one or more columns in a table for formatting.

3. Purpose / Function

Applies CSS styles (like column background colors or widths) to whole columns at once without repeating styles on every `<td>`.

4. Syntax

```
<colgroup>
  <col span="2" style="background-color: yellow;">
</colgroup>
```

5. Basic Example

```
<table>
  <colgroup>
    <col style="background-color: #f1f5f9;">
    <col style="background-color: #e2e8f0;">
  </colgroup>
  <tr><td>Col 1</td><td>Col 2</td></tr>
</table>
```

6. Output

Applies light grey background color to first column and darker grey to second column.

7. Attributes

Attribute	Description	Example	Required/Optional
<code>span</code>	Number of consecutive columns the group spans.	<code>span="3"</code>	Optional

8. Attribute Example

```
<colgroup span="2" class="col-highlight">
```

9. Real-World Example

- **Use Case:** Used in financial spreadsheets and comparison tables to highlight entire columns.
- **Code:**

```
<table>
  <colgroup>
    <col style="width: 20%;">
    <col style="width: 80%;">
  </colgroup>
  <tr><th>Label</th><th>Value</th></tr>
</table>
```

10. Important Notes

Must be placed inside `<table>`, after `<caption>` and before `<thead>` / `<tr>`. Contains `<col>` tags.

11. Related Tags

Nested in `<table>`. Contains `<col>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Not deprecated.

6.10 `<col>`

1. Tag Name

- **Tag Name:** `<col>`
- **Opening & Closing Form:** `<col ... >` (No closing tag)
- **Tag Type:** Empty / Void Tag

2. Definition

The `<col>` tag specifies column properties for each column within a `<colgroup>` element.

3. Purpose / Function

Sets column width or style properties across entire table columns.

4. Syntax

```
<col span="number" style="css_style">
```

5. Basic Example

```
<colgroup>
  <col style="width: 100px; background-color: #dbeafe;">
  <col style="width: 300px;">
</colgroup>
```

6. Output

Sets first table column to fixed 100px width with light blue fill.

7. Attributes

Attribute	Description	Example	Required/Optional
<code>span</code>	Specifies number of columns the <code><col></code> spans.	<code>span="2"</code>	Optional

8. Attribute Example

```
<col span="2" style="background-color: #fef08a;">
```

9. Real-World Example

- **Use Case:** Used in pricing grid columns to highlight recommended plan column.
- **Code:**

```
<colgroup>  
  <col span="1">  
  <col class="popular-column">  
</colgroup>
```

10. Important Notes

Does not hold content; used solely for styling column rules.

11. Related Tags

Nested inside `<colgroup>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Presentation attributes (`width` , `align` , `valign`) obsolete.

SECTION G: FORMS AND INPUT TAGS

7.1 `<form>`

1. Tag Name

- **Tag Name:** `<form>`
- **Opening & Closing Form:** `<form>...</form>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<form>` element represents a document section containing interactive controls for submitting user information to a web server.

3. Purpose / Function

Encapsulates inputs, buttons, checkboxes, dropdowns, and textareas to process user input data (e.g. login, registration, search).

4. Syntax

```
<form action="server_script.php" method="GET|POST" enctype="multipart/form-data">  
  <!-- Form inputs -->  
</form>
```

5. Basic Example

```
<form action="/submit-assignment" method="POST">  
  <label for="student">Student Name:</label>  
  <input type="text" id="student" name="student_name" required>  
  <button type="submit">Submit</button>  
</form>
```

6. Output

Renders a text box for student name and a submit button. Submitting sends data via POST request to `/submit-assignment`.

7. Attributes

Attribute	Description	Example	Required/Optional
<code>action</code>	Specifies URL endpoint where form data is sent on submit.	<code>action="/api/login"</code>	Recommended
<code>method</code>	HTTP method used (<code>GET</code> or <code>POST</code>).	<code>method="POST"</code>	Required
<code>enctype</code>	Specifies encoding type for form submission (<code>application/x-www-form-urlencoded</code> , <code>multipart/form-data</code> , <code>text/plain</code>).	<code>enctype="multipart/form-data"</code>	Required for file uploads
<code>target</code>	Window target for server response (<code>_self</code> , <code>_blank</code>).	<code>target="_blank"</code>	Optional
<code>autocomplete</code>	Enables/disables auto completion (<code>on</code> or <code>off</code>).	<code>autocomplete="on"</code>	Optional
<code>novalidate</code>	Disables default browser HTML5 validation checks when submitting.	<code>novalidate</code>	Optional
<code>name</code>	Specifies form name for script reference.	<code>name="loginForm"</code>	Optional

8. Attribute Example

```
<form action="upload.php" method="POST" enctype="multipart/form-data" autocomplete="off" novalid
```

9. Real-World Example

- **Use Case:** Used in every login box (Google, Facebook), sign up form, payment checkout, search bar, and feedback portal.
- **Code:**

```
<form action="/api/v1/register" method="POST" class="needs-validation">
  <h2>Student Registration Form</h2>
  <div class="mb-3">
    <label for="email">College Email</label>
    <input type="email" id="email" name="user_email" required class="form-control">
  </div>
  <button type="submit" class="btn btn-primary">Register Now</button>
</form>
```

10. Important Notes

Always specify `enctype="multipart/form-data"` on `<form>` when uploading files via `<input type="file">`; otherwise file binary data will not be transmitted.

11. Related Tags

Related to `<input>` , `<button>` , `<label>` , `<textarea>` , `<select>` .

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Not deprecated.

7.2 `<input>`

1. Tag Name

- **Tag Name:** `<input>`
- **Opening & Closing Form:** `<input ... >` (No closing tag)
- **Tag Type:** Empty / Void Tag

2. Definition

The `<input>` tag is used to create interactive controls for web-based forms to accept data from the user.

3. Purpose / Function

Serves as the primary versatile form element, rendering text boxes, checkboxes, radio buttons, file pickers, sliders, and submit controls depending on its `type` attribute value.

4. Syntax

```
<input type="type_name" name="field_name" value="initial_val">
```

5. Basic Example

```
<input type="text" name="username" placeholder="Enter USN Number" required>
```

6. Output

Renders a single-line text input field with gray placeholder text.

7. Attributes

Attribute	Description	Example	Required/Optional
<code>type</code>	Specifies input control type (<code>text</code> , <code>password</code> , <code>email</code> , <code>number</code> , etc.).	<code>type="email"</code>	Required
<code>name</code>	Name key submitted to server with field value.	<code>name="user_email"</code>	Mandatory for forms
<code>value</code>	Default or entered text value of input.	<code>value="Aryansh"</code>	Optional
<code>placeholder</code>	Short hint displayed inside empty field.	<code>placeholder="e.g. 1VU22CS001"</code>	Optional
<code>required</code>	Mandates field must be filled before submission.	<code>required</code>	Optional
<code>readonly</code>	Prevents user modification while retaining submit value.	<code>readonly</code>	Optional
<code>disabled</code>	Disables interaction and excludes from form submission.	<code>disabled</code>	Optional
<code>minlength</code>	Minimum character length required.	<code>minlength="8"</code>	Optional
<code>maxlength</code>	Maximum character length allowed.	<code>maxlength="50"</code>	Optional
<code>min</code>	Minimum numerical or date boundary value.	<code>min="0"</code>	Optional
<code>max</code>	Maximum numerical or date boundary value.	<code>max="100"</code>	Optional
<code>step</code>	Numerical stepping interval (<code>step="0.01"</code>).	<code>step="5"</code>	Optional
<code>pattern</code>	Regular expression regex string for value validation.	<code>pattern="[A-Z]{3}[0-9]{4}"</code>	Optional
<code>checked</code>	Specifies radio or checkbox is selected by default.	<code>checked</code>	Optional (radio/checkbox)
<code>multiple</code>	Allows selecting multiple values/files.	<code>multiple</code>	Optional (file/email)
<code>accept</code>	Specifies acceptable MIME file extensions.	<code>accept=".pdf,.png"</code>	Optional (file type)
<code>autocomplete</code>	Enables browser auto-fill suggestions.	<code>autocomplete="username"</code>	Optional

8. Attribute Example

```
<input type="password" id="pwd" name="user_pass" minlength="8" maxlength="20" required placeholder="Password" />
```

9. Real-World Example

- **Use Case:** Used in every web form across the internet.
- **Code:**

```
<!-- Real World Input Element with Validation -->
<input type="text"
  id="usn"
  name="student_usn"
  pattern="1[A-Z]{2}[0-9]{2}[A-Z]{2}[0-9]{3}"
  placeholder="1VU22CS001"
  title="Enter valid VTU USN Format"
  required
  class="form-input">
```

10. Important Notes

Inputs must always have a corresponding `<label>` tag for accessibility. Always include `name` attribute on `<input>`; unnamed inputs are ignored during form POST submissions.

11. Related Tags

Related to `<label>`, `<form>`, `<datalist>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Attributes `align`, `usemap` obsolete.

7.2.1 Detailed Explanation of Major `<input>` Types

Input Type	Description & Purpose	Syntax Example	HTML5 Validation Behavior
<code>text</code>	Standard single-line plain text box.	<code><input type="text" name="fname"></code>	General text string input.
<code>password</code>	Masks typed characters (bullets/asterisks) for security.	<code><input type="password" name="pwd"></code>	Security masking enabled.
<code>email</code>	Accepts email address format.	<code><input type="email" name="user_email"></code>	Validates <code>name@domain.com</code> format.
<code>number</code>	Accepts numeric values with spin arrows.	<code><input type="number" min="1" max="100" step="1"></code>	Restricts non-numeric characters.
<code>tel</code>	Input field for phone numbers.	<code><input type="tel" pattern="[0-9]{10}"></code>	Best combined with regex <code>pattern</code> .
<code>url</code>	Field for entering web addresses (URLs).	<code><input type="url" placeholder="https://"></code>	Validates <code>http://</code> or <code>https://</code> prefix.
<code>search</code>	Single-line search text box with clear (x) button.	<code><input type="search" name="query"></code>	Provides clear button on modern browsers.
<code>date</code>	Native date picker popup (Year, Month, Day).	<code><input type="date" name="dob"></code>	Format: <code>YYYY-MM-DD</code> .
<code>time</code>	Native time picker popup (Hours, Minutes).	<code><input type="time" name="alarm"></code>	Format: <code>HH:MM</code> (24-hr or AM/PM).
<code>datetime-local</code>	Combined date and time picker (No timezone).	<code><input type="datetime-local" name="event_time"></code>	Format: <code>YYYY-MM-DDTHH:MM</code> .
<code>month</code>	Native month and year selection control.	<code><input type="month" name="exp_month"></code>	Format: <code>YYYY-MM</code> .
<code>week</code>	Native week number and year selection.	<code><input type="week" name="proj_week"></code>	Format: <code>YYYY-Www</code> .
<code>color</code>	Native GUI color picker palette popup.	<code><input type="color" name="fav_color" value="#2563eb"></code>	Returns 6-digit Hex code (<code>#rrggbb</code>).
<code>checkbox</code>	Square toggle option allowing multiple selections.	<code><input type="checkbox" name="skills" value="html" checked></code>	Returns boolean/value if checked.

Input Type	Description & Purpose	Syntax Example	HTML5 Validation Behavior
radio	Circular mutually exclusive single-choice option button.	<code><input type="radio" name="gender" value="male"></code>	Grouped by identical <code>name</code> attribute.
file	File selection dialog picker for file uploads.	<code><input type="file" name="resume" accept=".pdf" multiple></code>	Requires <code>enctype="multipart/form-data"</code> .
hidden	Concealed input field carrying background data.	<code><input type="hidden" name="user_id" value="10928"></code>	Hidden from visual layout display.
range	Graphical slider bar for selecting approximate numeric value.	<code><input type="range" min="0" max="100" value="50"></code>	Default numeric slider UI control.
submit	Submit button that triggers form submission.	<code><input type="submit" value="Submit Form"></code>	Sends form payload to server.
reset	Reset button that clears all form inputs back to default values.	<code><input type="reset" value="Reset Fields"></code>	Clears all input field entries.
button	Generic clickable input button for JS scripts.	<code><input type="button" value="Click Me" onclick="doSomething()"></code>	No default submit behavior.

7.3 <label>

1. Tag Name

- **Tag Name:** `<label>`
- **Opening & Closing Form:** `<label>...</label>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<label>` element represents a caption for an item in a user interface, binding descriptive text to a form control element.

3. Purpose / Function

Improves usability by allowing users to click label text to focus the input box or toggle checkboxes, and enables screen readers to speak field captions.

4. Syntax

```
<label for="input_id">Field Label Text</label>
<input type="text" id="input_id">
```

5. Basic Example

```
<label for="phone">Phone Number:</label>
<input type="tel" id="phone" name="user_phone">
```

6. Output

Renders "Phone Number:" text next to input box. Clicking the text automatically activates and focuses the phone input box.

7. Attributes

Attribute	Description	Example	Required/Optional
<code>for</code>	Specifies the <code>id</code> of the target input element this label describes.	<code>for="user_email"</code>	Mandatory for explicit linkage

8. Attribute Example

```
<label for="terms">
  <input type="checkbox" id="terms" name="agree"> I agree to Terms & Conditions
</label>
```

9. Real-World Example

- **Use Case:** Used in every accessible web form across Google, Microsoft, and government portals.
- **Code:**

```
<div class="form-group">
  <label for="student_id" class="form-label">Student ID Card Number:</label>
  <input type="text" id="student_id" name="sid" class="form-control">
</div>
```

10. Important Notes

The `for` attribute on `<label>` MUST match the exact `id` attribute value of the associated input element.

11. Related Tags

Related to `<input>`, `<textarea>`, `<select>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Standard HTML5 accessibility element.

7.4 `<button>`

1. Tag Name

- **Tag Name:** `<button>`
- **Opening & Closing Form:** `<button>...</button>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<button>` element represents a clickable button, used to submit forms or execute custom JavaScript actions.

3. Purpose / Function

Renders styled clickable buttons that can contain rich HTML content like icons, formatted text, and images inside.

4. Syntax

```
<button type="submit|reset|button">Button Text</button>
```

5. Basic Example

```
<button type="submit" class="btn-submit">
  
  Send Assignment
</button>
```

6. Output

Renders a clickable button displaying an icon alongside the text "Send Assignment".

7. Attributes

Attribute	Description	Example	Required/Optional
<code>type</code>	Type behavior (<code>submit</code> , <code>reset</code> , <code>button</code>). Default in forms is <code>submit</code> .	<code>type="button"</code>	Strongly Recommended
<code>disabled</code>	Disables button interactions.	<code>disabled</code>	Optional
<code>name</code>	Name sent with form data on click.	<code>name="action"</code>	Optional
<code>value</code>	Value sent with form data on click.	<code>value="save_draft"</code>	Optional
<code>formaction</code>	Overrides parent form action URL on submit.	<code>formaction="/save-draft"</code>	Optional
<code>formmethod</code>	Overrides parent form HTTP method on submit.	<code>formmethod="POST"</code>	Optional

8. Attribute Example

```
<button type="button" onclick="window.print()" class="btn btn-outline-secondary">Print Hall Tick
```

9. Real-World Example

- **Use Case:** Used for submit buttons, modal dialog triggers, play/pause controls, and UI action triggers.
- **Code:**

```
<button type="submit" class="btn btn-primary btn-lg">
  <i class="fa fa-paper-plane"></i> Submit Registration
</button>
```

10. Important Notes

Always specify `type="button"` for non-submitting JS buttons inside forms; otherwise browser defaults to submitting the form (`type="submit"`).

11. Related Tags

Related to `<input type="button">`, `<form>`, `<a>` .

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Replaced `<input type="button">` due to richer styling support.

7.5 `<textarea>`

1. Tag Name

- **Tag Name:** `<textarea>`
- **Opening & Closing Form:** `<textarea>...</textarea>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<textarea>` tag defines a multi-line text input control for entering longer prose text.

3. Purpose / Function

Used for feedback forms, comment boxes, post composition, and multi-line address fields.

4. Syntax

```
<textarea name="field" rows="4" cols="50" placeholder="Hint"></textarea>
```

5. Basic Example

```
<textarea id="feedback" name="user_feedback" rows="5" cols="40" placeholder="Type your assignmen
```

6. Output

Renders a resizable multi-line text input box 5 rows tall and 40 columns wide.

7. Attributes

Attribute	Description	Example	Required/Optional
<code>name</code>	Name submitted with field content.	<code>name="comments"</code>	Required in forms
<code>rows</code>	Specifies visible height in text lines.	<code>rows="6"</code>	Recommended
<code>cols</code>	Specifies visible width in character average width.	<code>cols="50"</code>	Recommended
<code>placeholder</code>	Hint text inside empty textarea.	<code>placeholder="Enter text..."</code>	Optional
<code>maxlength</code>	Maximum allowed character count.	<code>maxlength="500"</code>	Optional
<code>required</code>	Mandates field completion.	<code>required</code>	Optional
<code>readonly</code>	Prevents user editing.	<code>readonly</code>	Optional
<code>wrap</code>	Text wrapping behavior (<code>soft</code> or <code>hard</code>).	<code>wrap="hard"</code>	Optional

8. Attribute Example

```
<textarea name="bio" rows="4" maxlength="300" placeholder="Brief student bio..." required></text
```

9. Real-World Example

- **Use Case:** Used in comment sections, blog article composition tools, review submission forms, and code playgrounds.
- **Code:**

```
<div class="form-group">
  <label for="address">Permanent Address:</label>
  <textarea id="address" name="address" rows="3" class="form-control" required></textarea>
</div>
```

10. Important Notes

Unlike `<input value="...">`, initial text for `<textarea>` is placed BETWEEN opening `<textarea>` and closing `</textarea>` tags.

11. Related Tags

Related to `<input>`, `<label>`, `<form>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Not deprecated.

7.6 <select>

1. Tag Name

- **Tag Name:** <select>
- **Opening & Closing Form:** <select>...</select>
- **Tag Type:** Paired / Container Tag

2. Definition

The <select> element is used to create a drop-down selection list of options.

3. Purpose / Function

Allows users to pick one or multiple options from a predefined drop-down list menu.

4. Syntax

```
<select name="select_name">
  <option value="val1">Option 1</option>
</select>
```

5. Basic Example

```
<label for="branch">Select Branch:</label>
<select id="branch" name="student_branch">
  <option value="cse">Computer Science</option>
  <option value="ece">Electronics & Communication</option>
  <option value="mech">Mechanical</option>
</select>
```

6. Output

Displays a drop-down menu defaulting to "Computer Science". Clicking opens list choices.

7. Attributes

Attribute	Description	Example	Required/Optional
name	Field name sent on form submit.	name="branch"	Required in forms
multiple	Allows selecting multiple options simultaneously (Ctrl/Cmd click).	multiple	Optional
size	Specifies visible dropdown rows without opening.	size="4"	Optional
required	Requires selection of non-empty option.	required	Optional
disabled	Disables dropdown interaction.	disabled	Optional

8. Attribute Example

```
<select name="subjects" id="sub" multiple size="3" class="form-select">
```

9. Real-World Example

- **Use Case:** Used for country select pickers, date dropdowns, state options, and branch selectors on university portals.
- **Code:**

```
<div class="mb-3">
  <label for="state">State of Residence:</label>
  <select id="state" name="state" required class="form-select">
    <option value="" selected disabled>-- Choose State --</option>
    <option value="KA">Karnataka</option>
    <option value="MH">Maharashtra</option>
  </select>
</div>
```

10. Important Notes

Direct child elements must be `<option>` or `<optgroup>` tags.

11. Related Tags

Related to `<option>`, `<optgroup>`, `<datalist>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Not deprecated.

7.7 `<option>`

1. Tag Name

- **Tag Name:** `<option>`
- **Opening & Closing Form:** `<option>...</option>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<option>` tag defines an item choice in a drop-down list created by `<select>`, `<optgroup>`, or `<datalist>`.

3. Purpose / Function

Specifies individual choices and values within selection menus.

4. Syntax

```
<option value="choice_value">Option Label</option>
```

5. Basic Example

```
<option value="sem3" selected>Semester 3</option>
```

6. Output

Renders "Semester 3" option selected by default in dropdown.

7. Attributes

Attribute	Description	Example	Required/Optional
value	Data payload sent to server when option selected.	value="IN"	Recommended
selected	Pre-selects this option by default when loaded.	selected	Optional
disabled	Disables selecting this individual option item.	disabled	Optional

8. Attribute Example

```
<option value="us" disabled>United States (Unavailable)</option>
```

9. Real-World Example

- **Use Case:** Used inside every select dropdown menu.
- **Code:**

```
<select name="gender">
  <option value="" disabled selected>Choose Gender</option>
  <option value="female">Female</option>
  <option value="male">Male</option>
</select>
```

10. Important Notes

If `value` attribute is omitted, server receives the raw text inside `<option>...</option>`.

11. Related Tags

Nested in `<select>`, `<optgroup>`, `<datalist>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Not deprecated.

7.8 `<optgroup>`

1. Tag Name

- **Tag Name:** `<optgroup>`
- **Opening & Closing Form:** `<optgroup label="...">...</optgroup>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<optgroup>` element creates sub-grouped category headings for `<option>` items inside a `<select>` dropdown menu.

3. Purpose / Function

Organizes long list options into distinct categorized groups.

4. Syntax

```
<select>
  <optgroup label="Group Title">
    <option value="v1">Opt 1</option>
  </optgroup>
</select>
```

5. Basic Example

```
<select name="electives">
  <optgroup label="CSE Electives">
    <option value="ai">Artificial Intelligence</option>
    <option value="ml">Machine Learning</option>
  </optgroup>
  <optgroup label="ECE Electives">
    <option value="vlsi">VLSI Design</option>
  </optgroup>
</select>
```

6. Output

Renders dropdown with non-selectable bold category headers ("CSE Electives", "ECE Electives") grouping options.

7. Attributes

Attribute	Description	Example	Required/Optional
<code>label</code>	Category group header title text.	<code>label="UG Courses"</code>	Mandatory
<code>disabled</code>	Disables all option items inside group.	<code>disabled</code>	Optional

8. Attribute Example

```
<optgroup label="PG Stream" disabled>
```

9. Real-World Example

- **Use Case:** Used in vehicle selectors (Category: SUV, Sedan), location pickers (Continent > Country), and branch electives.
- **Code:**

```
<select name="car_model">
  <optgroup label="Electric Vehicles">
    <option value="tesla_3">Tesla Model 3</option>
  </optgroup>
</select>
```

10. Important Notes

The `label` attribute is required to provide title for the sub-group category header.

11. Related Tags

Nested in `<select>`. Contains `<option>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Not deprecated.

7.9 `<fieldset>`

1. Tag Name

- **Tag Name:** `<fieldset>`
- **Opening & Closing Form:** `<fieldset>...</fieldset>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<fieldset>` element is used to group related input controls and labels within a web form, drawing a visible box frame around them.

3. Purpose / Function

Logically organizes form fields into distinct logical modules (e.g., Personal Info, Academic Details).

4. Syntax

```
<fieldset>
  <legend>Section Title</legend>
  <!-- Input fields -->
</fieldset>
```

5. Basic Example

```
<fieldset>
  <legend>Personal Information</legend>
  <label>First Name: <input type="text" name="fname"></label><br>
  <label>Last Name: <input type="text" name="lname"></label>
</fieldset>
```

6. Output

Renders a grey bordered box enclosing the inputs with "Personal Information" embedded into the top line border.

7. Attributes

Attribute	Description	Example	Required/Optional
<code>disabled</code>	Disables ALL child form inputs contained inside group.	<code>disabled</code>	Optional
<code>form</code>	Associates fieldset with a parent <code><form></code> ID.	<code>form="regForm"</code>	Optional
<code>name</code>	Name of fieldset group.	<code>name="personal_group"</code>	Optional

8. Attribute Example

```
<fieldset disabled id="payment-fields">
```

9. Real-World Example

- **Use Case:** Used in multi-step wizard forms, checkout payment sections, and profile editing screens.
- **Code:**

```
<fieldset class="border p-3 rounded">  
  <legend class="w-auto px-2">Account Security</legend>  
  <input type="password" placeholder="New Password">  
</fieldset>
```

10. Important Notes

Greatly assists accessibility screen readers by contextualizing grouped inputs.

11. Related Tags

Contains `<legend>`. Parent is `<form>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Not deprecated.

7.10 `<legend>`

1. Tag Name

- **Tag Name:** `<legend>`
- **Opening & Closing Form:** `<legend>...</legend>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<legend>` tag defines a caption title for the parent `<fieldset>` element.

3. Purpose / Function

Displays section headers embedded into top borders of fieldset containers.

4. Syntax

```
<legend>Group Caption Title</legend>
```

5. Basic Example

```
<legend>Payment Options</legend>
```

6. Output

Renders caption text embedded directly on top border line of fieldset.

7. Attributes

Attribute	Description	Example	Required/Optional
<code>align</code>	Sets alignment (<code>left</code> , <code>center</code> , <code>right</code>).	<code>align="center"</code>	Deprecated

8. Attribute Example

```
<legend class="font-bold text-primary">Emergency Contact</legend>
```

9. Real-World Example

- **Use Case:** Used inside form fieldsets on university and bank portals.
- **Code:**

```
<fieldset>
  <legend>Guardian Details</legend>
  <input type="text" name="guardian_name">
</fieldset>
```

10. Important Notes

MUST be the VERY FIRST child element inside a `<fieldset>` element.

11. Related Tags

Parent must be `<fieldset>` . Related to `<caption>` .

12. Deprecated/Obsolete Status

Standard HTML5 Tag. `align` attribute is deprecated.

7.11 `<datalist>`

1. Tag Name

- **Tag Name:** `<datalist>`
- **Opening & Closing Form:** `<datalist>...</datalist>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<datalist>` tag contains a set of `<option>` elements that represent recommended auto-complete options for an `<input>` element.

3. Purpose / Function

Provides an interactive auto-suggest autocomplete dropdown for standard single-line text inputs while still allowing custom typing.

4. Syntax

```
<input list="list_id">
<datalist id="list_id">
  <option value="Option 1">
</datalist>
```

5. Basic Example

```
<label for="browser">Browser:</label>
<input list="browsers" id="browser" name="browser">
<datalist id="browsers">
  <option value="Chrome">
  <option value="Firefox">
  <option value="Edge">
  <option value="Safari">
</datalist>
```

6. Output

Displays a standard text box that shows auto-suggest dropdown list when user clicks or types inside it.

7. Attributes

Attribute	Description	Example	Required/Optional
Global Attributes	Standard global attributes.	<code>id="city-list"</code>	Mandatory (<code>id</code> matches input <code>list</code>)

8. Attribute Example

```
<datalist id="courses">
  <option value="Computer Science Engineering">
</datalist>
```

9. Real-World Example

- **Use Case:** Used in search search bars, airport location search boxes, and college degree selector fields.
- **Code:**

```
<label for="city">Search City:</label>
<input list="cities" id="city" name="city" placeholder="Type city name">
<datalist id="cities">
  <option value="Bangalore">
  <option value="Belagavi">
  <option value="Delhi">
</datalist>
```

10. Important Notes

The `id` attribute of `<datalist>` MUST match the `list` attribute of the bound `<input>` element.

11. Related Tags

Paired with `<input list="...">`. Contains `<option>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Standard HTML5 element.

7.12 `<output>`

1. Tag Name

- **Tag Name:** `<output>`
- **Opening & Closing Form:** `<output>...</output>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<output>` element represents the result of a calculation performed by JavaScript or form script operations.

3. Purpose / Function

Provides a semantic element for displaying live calculation results (e.g. range slider value, shopping total calculation).

4. Syntax

```
<output name="result_name" for="input_ids">Initial output</output>
```

5. Basic Example

```
<form oninput="x.value=parseInt(a.value)+parseInt(b.value)">
  <input type="number" id="a" value="50"> +
  <input type="number" id="b" value="25"> =
  <output name="x" for="a b">75</output>
</form>
```

6. Output

Renders calculate total "75". Changing inputs recalculates and updates text dynamically.

7. Attributes

Attribute	Description	Example	Required/Optional
<code>for</code>	Space-separated list of IDs of input elements contributing to output.	<code>for="num1 num2"</code>	Recommended
<code>name</code>	Field name sent on submission.	<code>name="total"</code>	Optional
<code>form</code>	Parent form ID.	<code>form="calcForm"</code>	Optional

8. Attribute Example

```
<output id="volume-val" for="vol-range">50%</output>
```

9. Real-World Example

- **Use Case:** Used in loan EMI calculators, dynamic shopping cart calculators, and range slider value labels.
- **Code:**

```
<input type="range" id="rating" min="1" max="10" oninput="out.value=this.value">
Score: <output id="out" for="rating">5</output>/10
```

10. Important Notes

Essential for semantic accessibility when displaying dynamic mathematical outputs.

11. Related Tags

Related to `<input type="range">`, `<progress>`, `<meter>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Standard HTML5 element.

7.13 `<progress>`

1. Tag Name

- **Tag Name:** `<progress>`
- **Opening & Closing Form:** `<progress>...</progress>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<progress>` element represents the completion progress of a task, such as a file download, installation, or form submission.

3. Purpose / Function

Displays a dynamic horizontal progress bar representing task completion status percentage.

4. Syntax

```
<progress value="current_val" max="max_val"></progress>
```

5. Basic Example

```
<label for="file">Downloading Assignment PDF:</label>
<progress id="file" value="70" max="100"> 70% </progress>
```

6. Output

Renders a blue horizontal progress bar filled to 70% length.

7. Attributes

Attribute	Description	Example	Required/Optional
<code>value</code>	Current completed value of progress.	<code>value="45"</code>	Required (if determinate)
<code>max</code>	Maximum target value (default is 1.0).	<code>max="100"</code>	Optional

8. Attribute Example

```
<progress value="0.75" max="1.0" class="w-100"></progress>
```

9. Real-World Example

- **Use Case:** Used in file upload widgets (Dropbox, Google Drive), multi-step registration progress bars, and quiz timers.
- **Code:**

```
<div class="upload-status">
  <p>Uploading resume.pdf...</p>
  <progress value="80" max="100"></progress> 80%
</div>
```

10. Important Notes

Do not use `<progress>` to display disk space or gauge measurements; use `<meter>` for scalar range measurements.

11. Related Tags

Related to `<meter>`, `<output>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Standard HTML5 element.

7.14 `<meter>`

1. Tag Name

- **Tag Name:** `<meter>`
- **Opening & Closing Form:** `<meter>...</meter>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<meter>` element represents a scalar measurement within a known range, or a fractional value (e.g. disk usage, relevance match score, battery level).

3. Purpose / Function

Displays a gauge bar indicating values relative to low, high, and optimum ranges.

4. Syntax

```
<meter value="val" min="min" max="max" low="low" high="high" optimum="opt"></meter>
```

5. Basic Example

```
<label for="disk">Server Storage Used:</label>
<meter id="disk" value="85" min="0" max="100" low="30" high="80" optimum="20">85%</meter>
```

6. Output

Renders a gauge bar coloured yellow/red signaling high storage capacity warning.

7. Attributes

Attribute	Description	Example	Required/Optional
<code>value</code>	Current numeric value measured.	<code>value="60"</code>	Required
<code>min</code>	Minimum bound of range (default 0).	<code>min="0"</code>	Optional
<code>max</code>	Maximum bound of range (default 1.0).	<code>max="100"</code>	Optional
<code>low</code>	Upper boundary of low range.	<code>low="30"</code>	Optional
<code>high</code>	Lower boundary of high range.	<code>high="80"</code>	Optional
<code>optimum</code>	Optimal value point in range.	<code>optimum="15"</code>	Optional

8. Attribute Example

```
<meter value="9" min="0" max="10" low="3" high="7" optimum="9">9/10</meter>
```

9. Real-World Example

- **Use Case:** Used in password strength gauges, battery indicators, storage quota meters, and voter results.
- **Code:**

```
<p>Password Strength: <meter value="0.9" optimum="1.0"></meter> Strong</p>
```

10. Important Notes

`<meter>` is for static fractional scalar measurements; `<progress>` is for dynamic task completion.

11. Related Tags

Related to `<progress>`, `<output>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Standard HTML5 element.

SECTION H: SEMANTIC LAYOUT TAGS

8.1 <header>

1. Tag Name

- **Tag Name:** <header>
- **Opening & Closing Form:** <header>...</header>
- **Tag Type:** Paired / Container Tag

2. Definition

The <header> element represents introductory content, typically containing a group of introductory or navigational aids.

3. Purpose / Function

Encapsulates headings (<h1> - <h6>), site branding logo, search form, and main navigation links.

4. Syntax

```
<header>
  <h1>Site Title</h1>
  <nav>...</nav>
</header>
```

5. Basic Example

```
<header>
  
  <h1>Visvesvaraya Technological University</h1>
  <nav><a href="#home">Home</a></nav>
</header>
```

6. Output

Renders site header banner block containing logo, main title, and navigation bar.

7. Attributes

Attribute	Description	Example	Required/Optional
Global Attributes	Standard global attributes.	class="site-header"	Optional

8. Attribute Example

```
<header class="bg-primary text-white p-3 shadow">
```

9. Real-World Example

- **Use Case:** Used at top of every website layout (e.g. Amazon header bar, Wikipedia top banner).
- **Code:**

```
<header class="navbar navbar-dark bg-dark">
  <a class="navbar-brand" href="#">College Portal</a>
</header>
```

10. Important Notes

Can be used as page top header OR inside `<article>` and `<section>` as article headers.

11. Related Tags

Related to `<footer>`, `<nav>`, `<main>`, `<h1>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Standard HTML5 semantic element.

8.2 `<footer>`

1. Tag Name

- **Tag Name:** `<footer>`
- **Opening & Closing Form:** `<footer>...</footer>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<footer>` element represents a footer for its nearest sectioning content or sectioning root element.

3. Purpose / Function

Contains copyright notices, author contact details, site map links, privacy policy, and terms of service.

4. Syntax

```
<footer>
  <p>&copy; 2026 University. All rights reserved.</p>
</footer>
```

5. Basic Example

```
<footer>
  <p>&copy; 2026 Department of CSE. Contact: <address>info@college.edu</address></p>
</footer>
```

6. Output

Renders a block section at bottom of document containing copyright and contact info.

7. Attributes

Attribute	Description	Example	Required/Optional
Global Attributes	Standard global attributes.	<code>class="site-footer"</code>	Optional

8. Attribute Example

```
<footer class="bg-gray-900 text-gray-300 py-6">
```

9. Real-World Example

- **Use Case:** Used at bottom of every web page for footers.
- **Code:**

```
<footer>  
  <div class="footer-links">  
    <a href="/privacy">Privacy Policy</a> | <a href="/terms">Terms</a>  
  </div>  
</footer>
```

10. Important Notes

Contact information inside `<footer>` should be enclosed in an `<address>` tag.

11. Related Tags

Related to `<header>`, `<address>`, `<small>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Standard HTML5 semantic element.

8.3 `<main>`

1. Tag Name

- **Tag Name:** `<main>`
- **Opening & Closing Form:** `<main>...</main>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<main>` element represents the dominant content of the `<body>` of a document.

3. Purpose / Function

Encapsulates unique primary content of the page, excluding repetitive sidebars, headers, footers, and nav bars.

4. Syntax

```
<main>  
  <h1>Primary Page Topic</h1>  
  <p>Main content area...</p>  
</main>
```

5. Basic Example

```
<body>
  <header>Nav</header>
  <main>
    <h2>HTML Assignment Submission</h2>
    <p>Upload document below.</p>
  </main>
  <footer>Footer</footer>
</body>
```

6. Output

Defines main content body area for browser rendering and screen reader landmark focus.

7. Attributes

Attribute	Description	Example	Required/Optional
Global Attributes	Standard global attributes.	<code>id="main-content"</code>	Optional

8. Attribute Example

```
<main id="main-content" role="main" class="container mt-4">
```

9. Real-World Example

- **Use Case:** Used in every modern web app layout to isolate main content.
- **Code:**

```
<main>
  <article>
    <h1>Lesson 1: Introduction</h1>
  </article>
</main>
```

10. Important Notes

There MUST NOT be more than ONE visible `<main>` element in a document. Must not be descendant of `<header>`, `<footer>`, or `<nav>`.

11. Related Tags

Related to `<article>`, `<section>`, `<body>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Standard HTML5 landmark element.

8.4 `<section>`

1. Tag Name

- **Tag Name:** `<section>`
- **Opening & Closing Form:** `<section>...</section>`

- **Tag Type:** Paired / Container Tag

2. Definition

The `<section>` element represents a standalone generic section of a document, which typically has its own heading.

3. Purpose / Function

Groups related content logically into thematic chapters, tabs, or sections.

4. Syntax

```
<section>
  <h2>Section Title</h2>
  <p>Section content...</p>
</section>
```

5. Basic Example

```
<section id="syllabus">
  <h2>Course Syllabus</h2>
  <p>Module 1: HTML5 Tags</p>
</section>
```

6. Output

Renders thematic section with heading and paragraph.

7. Attributes

Attribute	Description	Example	Required/Optional
Global Attributes	Standard global attributes.	<code>id="about-us"</code>	Optional

8. Attribute Example

```
<section id="features" class="py-5 bg-light">
```

9. Real-World Example

- **Use Case:** Used on landing pages to break content into Features, Testimonials, Pricing, and Contact sections.
- **Code:**

```
<section class="pricing-table">
  <h2>Subscription Plans</h2>
</section>
```

10. Important Notes

Should typically contain a heading element (`<h1>` - `<h6>`). If container is purely for CSS styling, use `<div>` instead.

11. Related Tags

Related to `<article>`, `<div>`, `<main>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Standard HTML5 semantic tag.

8.5 `<article>`

1. Tag Name

- **Tag Name:** `<article>`
- **Opening & Closing Form:** `<article>...</article>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<article>` element represents a self-contained composition in a document that is intended to be independently distributable or reusable (e.g. blog post, news story, forum post).

3. Purpose / Function

Encapsulates independent content units that make sense on their own when RSS syndicated or shared.

4. Syntax

```
<article>
  <h2>Article Title</h2>
  <p>Article content...</p>
</article>
```

5. Basic Example

```
<article>
  <h2>Web Technologies Lab Assignment 1</h2>
  <p>Published on August 28, 2026 by CSE Dept.</p>
  <p>Complete all HTML tag exercises.</p>
</article>
```

6. Output

Renders standalone article content block.

7. Attributes

Attribute	Description	Example	Required/Optional
Global Attributes	Standard global attributes.	<code>class="blog-post"</code>	Optional

8. Attribute Example

```
<article class="card p-4 mb-4 shadow-sm">
```

9. Real-World Example

- **Use Case:** Used for news posts (BBC, CNN), blog entries, product cards (Amazon), and user comments.
- **Code:**

```
<article class="news-item">
  <header><h3>New Library Opened</h3></header>
  <p>The new campus library opens today.</p>
</article>
```

10. Important Notes

An `<article>` should make complete sense independently if pulled out of the page layout.

11. Related Tags

Related to `<section>`, `<aside>`, `<header>`, `<footer>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Standard HTML5 semantic tag.

8.6 `<aside>`

1. Tag Name

- **Tag Name:** `<aside>`
- **Opening & Closing Form:** `<aside>...</aside>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<aside>` element represents a portion of a document whose content is only indirectly related to the document's main content (e.g. sidebars, callout boxes, ads, related links).

3. Purpose / Function

Renders peripheral content like sidebars, callouts, related article lists, or advertising blocks.

4. Syntax

```
<aside>
  <h3>Related Links</h3>
  <!-- Sidebar links -->
</aside>
```

5. Basic Example

```
<main><article>Main Lesson Content</article></main>
<aside>
  <h3>Quick Facts</h3>
  <p>HTML was created by Tim Berners-Lee in 1991.</p>
</aside>
```

6. Output

Renders sidebar callout content box adjacent to main article.

7. Attributes

Attribute	Description	Example	Required/Optional
Global Attributes	Standard global attributes.	<code>class="sidebar"</code>	Optional

8. Attribute Example

```
<aside class="col-md-4 sidebar-box">
```

9. Real-World Example

- **Use Case:** Used for blog sidebars, related posts boxes, ad banners, and Wikipedia info callouts.
- **Code:**

```
<aside class="widget-area">
  <div class="widget"><h4>Search Portal</h4></div>
</aside>
```

10. Important Notes

Do not put main article content inside `<aside>`; use only for auxiliary related sub-content.

11. Related Tags

Related to `<main>`, `<article>`, `<section>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Standard HTML5 semantic tag.

8.7 `<div>`

1. Tag Name

- **Tag Name:** `<div>`
- **Opening & Closing Form:** `<div>...</div>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<div>` (Content Division) element is the generic container for flow content. It has no effect on content or layout until styled using CSS.

3. Purpose / Function

Used as a non-semantic block container to group elements for CSS styling, Flexbox/Grid layout, or JavaScript DOM manipulation.

4. Syntax

```
<div class="wrapper">
  <!-- Child elements -->
</div>
```

5. Basic Example

```
<div class="card">
  <h3>Card Header</h3>
  <p>Card description text.</p>
</div>
```

6. Output

Renders a block box container styled according to CSS `.card` rules.

7. Attributes

Attribute	Description	Example	Required/Optional
Global Attributes	Standard global attributes (<code>id</code> , <code>class</code> , <code>style</code>).	<code>class="container"</code>	Optional

8. Attribute Example

```
<div id="app-root" class="d-flex flex-column align-items-center">
```

9. Real-World Example

- **Use Case:** Used extensively across all websites for layout wrappers, flex containers, modal boxes, grid rows, and UI components.
- **Code:**

```
<div class="row">
  <div class="col-6">Column 1</div>
  <div class="col-6">Column 2</div>
</div>
```

10. Important Notes

Use `<div>` ONLY when no semantic tag (`<article>` , `<section>` , `<header>` , `<main>`) is appropriate.

11. Related Tags

Related to `<span>` , `<section>` , `<article>` .

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Standard non-semantic block element.

8.8 `<span>`

1. Tag Name

- **Tag Name:** `<span>`
- **Opening & Closing Form:** `<span>...</span>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<span>` element is a generic inline container for phrasing content, which does not inherently represent anything.

3. Purpose / Function

Used to group inline elements or text fragments for CSS styling (colors, fonts) or JavaScript manipulation without starting a new line.

4. Syntax

```
<p>Text <span class="highlight">styled text</span> text continuation.</p>
```

5. Basic Example

```
<p>Status: <span style="color: green; font-weight: bold;">APPROVED</span></p>
```

6. Output

Renders word "APPROVED" in bold green text inline within paragraph.

7. Attributes

Attribute	Description	Example	Required/Optional
Global Attributes	Standard global attributes.	<code>class="badge"</code>	Optional

8. Attribute Example

```
<span class="badge bg-success text-white">Passed</span>
```

9. Real-World Example

- **Use Case:** Used for badge tags, colored status text, icon wrappers, price highlights, and inline JS hooks.
- **Code:**

```
<p>Price: <span class="price-tag">$49.99</span></p>
```

10. Important Notes

`<span>` is an INLINE element. It does not force newlines. Use `<div>` for block containers.

11. Related Tags

Related to `<div>`, `<mark>`, `<strong>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Standard non-semantic inline element.

SECTION I: INTERACTIVE ELEMENTS

9.1 `<details>`

1. Tag Name

- **Tag Name:** `<details>`
- **Opening & Closing Form:** `<details>...</details>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<details>` tag creates an interactive disclosure widget that the user can open and close to reveal or hide additional information.

3. Purpose / Function

Implements native HTML accordion dropdowns and collapsible FAQ panels without requiring JavaScript.

4. Syntax

```
<details open>
  <summary>Click to expand</summary>
  <p>Hidden content details...</p>
</details>
```

5. Basic Example

```
<details>
  <summary>View Assignment Guidelines</summary>
  <p>1. Must be submitted in PDF format.</p>
  <p>2. Maximum length 10 pages.</p>
</details>
```

6. Output

Renders a clickable arrow toggle titled "View Assignment Guidelines". Clicking expands and reveals the hidden guidelines.

7. Attributes

Attribute	Description	Example	Required/Optional
<code>open</code>	Specifies that details should be expanded/visible by default on load.	<code>open</code>	Optional

8. Attribute Example

```
<details open class="faq-item">
```

9. Real-World Example

- **Use Case:** Used in FAQ accordion lists, spoiler warnings, GitHub pull request details, and expandable sidebar menus.

- **Code:**

```
<details>
  <summary>Question 1: What is HTML5?</summary>
  <p>HTML5 is the latest version of HyperText Markup Language.</p>
</details>
```

10. Important Notes

Pair with `<summary>` as first child tag. Clicking summary toggles `open` attribute state automatically.

11. Related Tags

Contains `<summary>`. Related to `<dialog>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Standard HTML5 interactive element.

9.2 `<summary>`

1. Tag Name

- **Tag Name:** `<summary>`
- **Opening & Closing Form:** `<summary>...</summary>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<summary>` element specifies a visible heading or legend for a parent `<details>` disclosure box.

3. Purpose / Function

Acts as the clickable header text for expanding/collapsing `<details>` widgets.

4. Syntax

```
<summary>Clickable Heading Text</summary>
```

5. Basic Example

```
<summary>Click here to view hints</summary>
```

6. Output

Renders clickable heading text with small triangular expand arrow next to it.

7. Attributes

Attribute	Description	Example	Required/Optional
Global Attributes	Standard global attributes.	<code>class="fw-bold"</code>	Optional

8. Attribute Example

```
<summary class="font-weight-bold text-primary">Module 1 FAQ</summary>
```

9. Real-World Example

- **Use Case:** Used as header title in accordions across documentation platforms (GitHub, MDN).
- **Code:**

```
<details>
  <summary>Click for Solution Code</summary>
  <pre><code>cout &lt;&lt; "Solved!";</code></pre>
</details>
```

10. Important Notes

Must be the VERY FIRST child element inside a `<details>` element.

11. Related Tags

Parent element must be `<details>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Standard HTML5 element.

9.3 `<dialog>`

1. Tag Name

- **Tag Name:** `<dialog>`
- **Opening & Closing Form:** `<dialog>...</dialog>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<dialog>` tag defines a native modal dialog window or popup box in an HTML document.

3. Purpose / Function

Implements native popups, alert modals, confirmation dialogs, and popup submission forms.

4. Syntax

```
<dialog id="myModal" open>
  <p>Modal Dialog Window</p>
  <button onclick="myModal.close()">Close</button>
</dialog>
```

5. Basic Example

```
<dialog id="confirmDialog">
  <h3>Confirm Submission</h3>
  <p>Are you sure you want to submit?</p>
  <button onclick="document.getElementById('confirmDialog').close()">Cancel</button>
</dialog>
```

6. Output

Renders modal overlay dialog window on page when activated via JS `.showModal()`.

7. Attributes

Attribute	Description	Example	Required/Optional
<code>open</code>	Indicates modal is currently active and visible.	<code>open</code>	Optional

8. Attribute Example

```
<dialog id="loginModal" class="p-4 rounded shadow-lg">
```

9. Real-World Example

- **Use Case:** Used for web app login popups, cookie consent popups, and confirmation dialog boxes.
- **Code:**

```
<dialog id="termsModal">
  <form method="dialog">
    <p>Agree to terms?</p>
    <button value="cancel">Cancel</button>
    <button value="default">Agree</button>
  </form>
</dialog>
```

10. Important Notes

Use JavaScript `.showModal()` method to open as backdrop modal, and `.close()` method to dismiss.

11. Related Tags

Related to `<details>`, `<form>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Standard HTML5 element.

SECTION J: SCRIPTING AND EMBEDDED CONTENT

10.1 `<script>`

1. Tag Name

- **Tag Name:** `<script>`
- **Opening & Closing Form:** `<script>...</script>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<script>` element is used to embed executable JavaScript code or reference external JavaScript files via the `src` attribute.

3. Purpose / Function

Adds dynamic interactivity, DOM manipulation, form validation, and API fetch calls to web pages.

4. Syntax

```
<script src="app.js" async defer></script>
<!-- OR Inline Script -->
<script>
  // JS code
</script>
```

5. Basic Example

```
<script>
  function showGreeting() {
    alert("Welcome to HTML Web Tech Assignment!");
  }
</script>
```

6. Output

Executes JavaScript logic in browser engine without displaying visual content text on body.

7. Attributes

Attribute	Description	Example	Required/Optional
<code>src</code>	Path to external JavaScript file (.js).	<code>src="main.js"</code>	Required for external scripts
<code>async</code>	Downloads script asynchronously and executes immediately when loaded.	<code>async</code>	Optional
<code>defer</code>	Downloads script asynchronously and executes only AFTER HTML parsing finishes.	<code>defer</code>	Recommended
<code>type</code>	MIME type or module definition (<code>text/javascript</code> , <code>module</code>).	<code>type="module"</code>	Optional
<code>crossorigin</code>	Configures CORS requests for external scripts.	<code>crossorigin="anonymous"</code>	Optional

8. Attribute Example

```
<script src="https://cdn.example.com/app.js" defer type="module"></script>
```

9. Real-World Example

- **Use Case:** Used on every single web application to include frameworks (React, Vue, jQuery) and interactive code.
- **Code:**

```
<script src="bootstrap.bundle.min.js" defer></script>
```

10. Important Notes

Place external script links in `<head>` with the `defer` attribute to avoid blocking page rendering HTML parsing.

11. Related Tags

Related to `<noscript>`, `<template>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. `language` attribute is obsolete.

10.2 `<noscript>`

1. Tag Name

- **Tag Name:** `<noscript>`
- **Opening & Closing Form:** `<noscript>...</noscript>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<noscript>` element defines alternative content to be rendered for users who have disabled JavaScript in their browser or have a browser that doesn't support scripts.

3. Purpose / Function

Provides fallback warning messages when JavaScript is required to view a web application.

4. Syntax

```
<noscript>
  <p>Warning: JavaScript is disabled in your browser.</p>
</noscript>
```

5. Basic Example

```
<noscript>
  <div style="color: red; padding: 10px;">
    JavaScript is turned off! Please enable JavaScript to submit this assignment.
  </div>
</noscript>
```

6. Output

Renders warning banner ONLY if browser JavaScript execution is disabled.

7. Attributes

Attribute	Description	Example	Required/Optional
Global Attributes	Standard global attributes.	<code>class="alert-box"</code>	Optional

8. Attribute Example

```
<noscript class="bg-warning text-dark p-3">
```

9. Real-World Example

- **Use Case:** Used in Single Page Applications (React, Angular, Next.js) to notify users if JS runtime is disabled.
- **Code:**

```
<noscript>
  <h2>JavaScript Required</h2>
  <p>We're sorry but this app doesn't work properly without JavaScript enabled.</p>
</noscript>
```

10. Important Notes

Content inside `<noscript>` is completely ignored by browser engine if JavaScript is enabled.

11. Related Tags

Related to `<script>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Standard HTML5 fallback tag.

10.3 <canvas>

1. Tag Name

- **Tag Name:** <canvas>
- **Opening & Closing Form:** <canvas>...</canvas>
- **Tag Type:** Paired / Container Tag

2. Definition

The <canvas> tag is used to draw graphics on the fly via JavaScript scripting (2D drawing context or 3D WebGL context).

3. Purpose / Function

Renders dynamic interactive graphs, charts, 2D/3D browser games, image processing filters, and animations.

4. Syntax

```
<canvas id="myCanvas" width="400" height="200">Fallback text</canvas>
```

5. Basic Example

```
<canvas id="graphCanvas" width="500" height="300" style="border:1px solid #ccc;"></canvas>
<script>
  const ctx = document.getElementById("graphCanvas").getContext("2d");
  ctx.fillStyle = "blue";
  ctx.fillRect(10, 10, 150, 80);
</script>
```

6. Output

Renders a 500x300 pixel graphics box containing a blue rectangle.

7. Attributes

Attribute	Description	Example	Required/Optional
width	Specifies canvas coordinate grid width in pixels (default 300).	width="800"	Recommended
height	Specifies canvas coordinate grid height in pixels (default 150).	height="600"	Recommended

8. Attribute Example

```
<canvas id="gameCanvas" width="800" height="600"></canvas>
```

9. Real-World Example

- **Use Case:** Used in Chart.js data graphs, browser games (Phaser), Three.js 3D web graphics, and photo editors (Figma canvas).
- **Code:**

```
<canvas id="signature-pad" width="400" height="200" class="border"></canvas>
```

10. Important Notes

Must set `width` and `height` as direct attributes on `<canvas>`, NOT in CSS, to prevent coordinate resolution distortion.

11. Related Tags

Related to `<script>`, `<img>`, `<svg>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Standard HTML5 graphics element.

10.4 `<template>`

1. Tag Name

- **Tag Name:** `<template>`
- **Opening & Closing Form:** `<template>...</template>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<template>` tag is a mechanism for holding HTML content that is not rendered when the page loads, but can be instantiated later at runtime using JavaScript.

3. Purpose / Function

Stores reusable HTML markup blueprints (e.g., dynamic table row templates, card templates) to clone via JS DOM scripts.

4. Syntax

```
<template id="my-tpl">
  <!-- HTML blueprint -->
</template>
```

5. Basic Example

```
<template id="row-template">
  <tr>
    <td class="usn"></td>
    <td class="name"></td>
  </tr>
</template>
```

6. Output

Hidden on initial load. Content inside `<template>` is invisible until cloned and inserted into DOM via JavaScript.

7. Attributes

Attribute	Description	Example	Required/Optional
Global Attributes	Standard global attributes (<code>id</code>).	<code>id="card-tp1"</code>	Mandatory (<code>id</code> for JS reference)

8. Attribute Example

```
<template id="comment-template">
```

9. Real-World Example

- **Use Case:** Used in modern Web Components framework architectures, dynamic table generators, and SPA rendering.
- **Code:**

```
<template id="item-template">  
  <li class="todo-item"><span class="title"></span></li>  
</template>
```

10. Important Notes

Browser parses content inside `<template>` for syntax errors on load, but does not render images or run scripts inside it until instantiated.

11. Related Tags

Related to `<slot>`, `<script>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Standard HTML5 Web Component tag.

10.5 `<slot>`

1. Tag Name

- **Tag Name:** `<slot>`
- **Opening & Closing Form:** `<slot>...</slot>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<slot>` element — part of the Web Components technology suite — is a placeholder inside a Web Component shadow DOM that you can fill with your own markup.

3. Purpose / Function

Enables component slot composition in custom Shadow DOM elements.

4. Syntax

```
<slot name="slot_name">Default fallback text</slot>
```

5. Basic Example

```
<template id="my-card">
  <h2><slot name="card-title">Default Title</slot></h2>
  <p><slot name="card-body">Default Body</slot></p>
</template>
```

6. Output

Renders custom slotted content passed into Web Component instance.

7. Attributes

Attribute	Description	Example	Required/Optional
name	Specifies slot identifier name for target insertion.	name="header"	Optional

8. Attribute Example

```
<slot name="user-name">Guest User</slot>
```

9. Real-World Example

- **Use Case:** Used in custom Web Component UI libraries (LitElement, Stencil, Shoelace).
- **Code:**

```
<custom-element>
  <span slot="user-name">Aryansh</span>
</custom-element>
```

10. Important Notes

Used specifically inside Web Components Shadow DOM structures.

11. Related Tags

Related to `<template>`, `<script>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Standard HTML5 Web Component tag.

SECTION K: TEXT EDITING AND RUBY ELEMENTS

11.1 <time>

1. Tag Name

- **Tag Name:** <time>
- **Opening & Closing Form:** <time>...</time>
- **Tag Type:** Paired / Container Tag

2. Definition

The <time> element represents a specific period in time or a date.

3. Purpose / Function

Provides machine-readable date/time annotations for search engines, calendar integrations, and localized date formatting.

4. Syntax

```
<time datetime="YYYY-MM-DDThh:mm:ss">Visible Date Text</time>
```

5. Basic Example

```
<p>Assignment due date: <time datetime="2026-08-31T23:59">August 31, 2026 at midnight</time></p>
```

6. Output

Displays "August 31, 2026 at midnight" while providing ISO machine format 2026-08-31T23:59 to search engines.

7. Attributes

Attribute	Description	Example	Required/Optional
datetime	Machine-readable date/time string in ISO 8601 format.	datetime="2026-08-28"	Mandatory for machine parsing

8. Attribute Example

```
<time datetime="2026-08-28T20:50:00+05:30">August 28, 2026</time>
```

9. Real-World Example

- **Use Case:** Used on news articles (published dates), event management sites, blog timestamps, and calendar invites.
- **Code:**

```
<article>
  <p>Posted on <time datetime="2026-08-28">Today</time></p>
</article>
```

10. Important Notes

Always provide standardized ISO format (YYYY-MM-DD) in `datetime` attribute.

11. Related Tags

Related to `<data>`, `<mark>` .

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Standard HTML5 semantic tag.

11.2 `<data>`

1. Tag Name

- **Tag Name:** `<data>`
- **Opening & Closing Form:** `<data>...</data>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<data>` element links a given piece of content with a machine-readable translation value.

3. Purpose / Function

Associates plain text values with machine-readable product SKUs, currency amounts, or numerical IDs.

4. Syntax

```
<data value="machine_val">Human visible text</data>
```

5. Basic Example

```
<p>Product: <data value="SKU-98745">Wireless Mouse</data> - Price: $25</p>
```

6. Output

Renders "Wireless Mouse" to user while linking background machine SKU `SKU-98745` .

7. Attributes

Attribute	Description	Example	Required/Optional
<code>value</code>	Machine-readable data value string.	<code>value="10928"</code>	Required

8. Attribute Example

```
<data value="978-0131103627">C Programming Language Book</data>
```

9. Real-World Example

- **Use Case:** Used in eCommerce product catalogs, inventory management, and database record views.
- **Code:**

```
<ul>
  <li><data value="PROD-01">Laptop</data></li>
</ul>
```

10. Important Notes

Use `<time>` for dates/times and `<data>` for generic machine-readable strings/numbers.

11. Related Tags

Related to `<time>`, `<meter>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Standard HTML5 element.

11.3 `<ruby>`

1. Tag Name

- **Tag Name:** `<ruby>`
- **Opening & Closing Form:** `<ruby>...</ruby>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<ruby>` element represents small annotations renderable above or next to base text, used in East Asian typography (Furigana annotations).

3. Purpose / Function

Provides pronunciation guides for East Asian characters (Japanese Kanji, Chinese Hanzi).

4. Syntax

```
<ruby> Base Text <rt> Annotation </rt> </ruby>
```

5. Basic Example

```
<ruby> 漢 <rt> かん </rt> 字 <rt> じ </rt> </ruby>
```

6. Output

Displays Japanese Kanji characters with small phonetic Hiragana guide reading annotations above them.

7. Attributes

Attribute	Description	Example	Required/Optional
Global Attributes	Standard global attributes.	<code>lang="ja"</code>	Optional

8. Attribute Example

```
<ruby lang="ja"> 明日 <rt> あした </rt> </ruby>
```

9. Real-World Example

- **Use Case:** Used in Japanese and Chinese language learning websites and East Asian news portals.
- **Code:**

```
<p><ruby> 東京 <rp>(</rp><rt>とうきょう</rt><rp>)</rp> </ruby></p>
```

10. Important Notes

Contains `<rt>` (ruby text) and optional `<rp>` (ruby parenthesis fallback).

11. Related Tags

Contains `<rt>` and `<rp>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Standard HTML5 international typography element.

11.4 `<rt>`

1. Tag Name

- **Tag Name:** `<rt>`
- **Opening & Closing Form:** `<rt>...</rt>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<rt>` tag marks the ruby text component of a ruby annotation, providing pronunciation or translation subtext.

3. Purpose / Function

Renders small phonetic sub-text above base character in `<ruby>`.

4. Syntax

```
<rt>Phonetic Reading</rt>
```

5. Basic Example

```
<ruby>字<rt>じ</rt></ruby>
```

6. Output

Renders small text "じ" above "字".

7. Attributes

Attribute	Description	Example	Required/Optional
Global Attributes	Standard global attributes.	<code>class="furigana"</code>	Optional

8. Attribute Example

```
<rt class="small-text">reading</rt>
```

9. Real-World Example

- **Use Case:** Used inside `<ruby>` tags.
- **Code:**

```
<ruby>本<rt>ほん</rt></ruby>
```

10. Important Notes

Must be nested inside a `<ruby>` element.

11. Related Tags

Parent is `<ruby>`. Sibling is `<rp>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Standard HTML5 element.

11.5 `<rp>`

1. Tag Name

- **Tag Name:** `<rp>`
- **Opening & Closing Form:** `<rp>...</rp>`
- **Tag Type:** Paired / Container Tag

2. Definition

The `<rp>` (Ruby Parentheses) tag provides fallback parentheses to surround ruby text for browsers that do not support `<ruby>` annotations.

3. Purpose / Function

Provides fallback display (reading) on non-ruby legacy browsers.

4. Syntax

```
<rp>(</rp><rt>text</rt><rp>)</rp>
```

5. Basic Example

```
<ruby>漢<rp>(</rp><rt>かん</rt><rp>)</rp></ruby>
```

6. Output

Modern browsers hide `<rp>` content. Legacy non-ruby browsers display fallback: 漢(かん).

7. Attributes

Attribute	Description	Example	Required/Optional
Global Attributes	Standard global attributes.	<code>class="fallback-parens"</code>	Optional

8. Attribute Example

```
<rp></rp>
```

9. Real-World Example

- **Use Case:** Used for backward-compatible East Asian site localization.
- **Code:**

```
<ruby>日<rp>[</rp><rt>に</rt><rp>]</rp></ruby>
```

10. Important Notes

Nested in `<ruby>` immediately before and after `<rt>`.

11. Related Tags

Nested in `<ruby>`. Sibling to `<rt>`.

12. Deprecated/Obsolete Status

Standard HTML5 Tag. Standard HTML5 element.

SECTION L: DEPRECATED AND OBSOLETE TAGS

The following tags were used in older HTML versions (HTML 4.01 / XHTML) for presentation and styling, but are **OFFICIALLY OBSOLETE / DEPRECATED in HTML5**. Modern web developers **MUST NOT** use these tags; use CSS for styling instead.

12.1

- **Status: OBSOLETE / DEPRECATED**
- **Reason:** Violated separation of concerns by mixing presentation with HTML structure.
- **Obsolete Syntax:** `<font size="4" color="red" face="Arial">Text</font>`
- **Modern HTML5 + CSS Alternative:** Use CSS properties: `font-size`, `color`, and `font-family`.

```
<!-- Modern Alternative -->
<span style="font-size: 18px; color: red; font-family: Arial, sans-serif;">Styled Text</span>
```

12.2 <center>

- **Status: OBSOLETE / DEPRECATED**
- **Reason:** Presentation-only tag used to center text and elements horizontally.
- **Obsolete Syntax:** `<center><p>Centered Content</p></center>`
- **Modern HTML5 + CSS Alternative:** Use CSS `text-align: center;` or Flexbox `justify-content: center;`.

```
<!-- Modern Alternative -->
<div style="text-align: center;">Centered Content</div>
```

12.3 <marquee>

- **Status: OBSOLETE / DEPRECATED**
- **Reason:** Non-standard tag that created scrolling text animations. It harms web accessibility (WCAG) and causes usability distractions.
- **Obsolete Syntax:** `<marquee direction="left">Scrolling News Text</marquee>`
- **Modern HTML5 + CSS Alternative:** Use CSS3 `@keyframes` animations or JavaScript marquee libraries.

```
<!-- Modern CSS Alternative -->
<div class="scrolling-container">
  <p class="animate-marquee">Modern Accessible Scrolling Text</p>
</div>
```

12.4 <big>

- **Status: OBSOLETE / DEPRECATED**
- **Reason:** Presentation tag that increased font size by one size relative to surrounding text.

- **Obsolete Syntax:** `<big>Larger Text</big>`
- **Modern HTML5 + CSS Alternative:** Use CSS `font-size: larger;` or `font-size: 1.25rem;` .

```
<!-- Modern Alternative -->
<span style="font-size: 1.25em;">Larger Text</span>
```

12.5 `<strike>`

- **Status:** OBSOLETE / DEPRECATED
- **Reason:** Purely visual strikethrough tag without semantic meaning.
- **Obsolete Syntax:** `<strike>Strikethrough text</strike>`
- **Modern HTML5 + CSS Alternative:** Use `<del>` (for deleted content edits) or `<s>` (for non-accurate content), or CSS `text-decoration: line-through;` .

```
<!-- Modern Alternative -->
<del>Deleted text revision</del>
```

12.6 `<tt>` (Teletype Text)

- **Status:** OBSOLETE / DEPRECATED
- **Reason:** Legacy tag used for rendering fixed-width teletype font.
- **Obsolete Syntax:** `<tt>Monospace text</tt>`
- **Modern HTML5 + CSS Alternative:** Use `<code>` , `<kbd>` , `<samp>` , or CSS `font-family: monospace;` .

```
<!-- Modern Alternative -->
<code>Monospace Code Text</code>
```

12.7 `<frame>` and `<frameset>`

- **Status:** OBSOLETE / DEPRECATED
- **Reason:** Divided browser screen into multiple frame sub-windows. Broken bookmarks, print layouts, SEO indexing, and mobile accessibility completely.
- **Obsolete Syntax:** `<frameset cols="50%,50%"><frame src="page1.html"><frame src="page2.html"></frameset>`
- **Modern HTML5 + CSS Alternative:** Use standard responsive layout `<div>` containers styled with CSS Flexbox or CSS Grid, or embedded `<iframe>` tags.

ADDITIONAL COMPREHENSIVE STUDY SECTIONS

1. HTML Global Attributes

Global attributes are attributes that can be used on **almost any HTML element**, regardless of its category or specific tag type.

Global Attributes Reference Table

Global Attribute	Purpose / Function	Example Usage
<code>id</code>	Specifies a unique identifier for an element in the entire DOM tree. Used for CSS styling, JS DOM selection, and URL fragment anchor links.	<code><div id="header-nav"></code>
<code>class</code>	Specifies one or more class names for an element. Used to group multiple elements for CSS styling and JS selection.	<code><p class="lead text-primary"></code>
<code>style</code>	Allows applying inline CSS styling rules directly to an element.	<code><h1 style="color: blue; margin-top: 10px;"></code>
<code>title</code>	Provides advisory tooltip information that appears when hovering over the element with a mouse.	<code><abbr title="HyperText Markup Language">HTML</abbr></code>
<code>lang</code>	Specifies the language code of the element's content.	<code><p lang="fr">Bonjour le monde</p></code>
<code>dir</code>	Specifies text directionality (<code>ltr</code> for Left-To-Right, <code>rtl</code> for Right-To-Left, <code>auto</code> for browser choice).	<code><p dir="rtl">سلام</p></code>
<code>hidden</code>	Boolean attribute that visually hides the element from display on the page.	<code><div hidden>Hidden content</div></code>
<code>tabindex</code>	Controls keyboard tab navigation order and focusability (<code>0</code> puts in normal tab order, <code>-1</code> focusable via JS only, <code>>0</code> explicit tab index order).	<code><div tabindex="0" role="button"></code>
<code>contenteditable</code>	Boolean/enum attribute (<code>true</code> / <code>false</code>) specifying whether element content is editable by the user.	<code><div contenteditable="true">Editable note</div></code>
<code>draggable</code>	Specifies whether an element is draggable using Drag and Drop APIs (<code>true</code> , <code>false</code> , <code>auto</code>).	<code></code>
<code>spellcheck</code>	Specifies whether element text should be checked for spelling/grammar errors by browser engine (<code>true</code> / <code>false</code>).	<code><textarea spellcheck="true"></code>
<code>translate</code>	Hints whether element content should be translated when page localization occurs (<code>yes</code> / <code>no</code>).	<code>BrandName</code>
<code>data-*</code>	Custom data attributes allowing developers to store custom data payload attributes on HTML elements.	<code><button data-user-id="1092" data-role="admin"></code>

Global Attribute	Purpose / Function	Example Usage
<code>role</code>	WAI-ARIA role attribute defining element semantic purpose for assistive technologies.	<code><div role="navigation" aria-label="Main"></code>
<code>accesskey</code>	Specifies a keyboard shortcut key to activate or focus the element.	<code><button accesskey="s">Save (Alt+S) </button></code>

2. Event Handler Attributes

Event handler attributes allow executing inline JavaScript functions directly when specific user interactions or system events occur on HTML elements.

Common Event Handler Attributes Table

Event Attribute	Trigger Condition / Description	Example Usage
<code>onclick</code>	Fires when user clicks mouse primary button on element.	<code><button onclick="alert('Clicked!')">Click</button></code>
<code>ondblclick</code>	Fires when user double-clicks element.	<code><div ondblclick="zoomImage()">Double Click</div></code>
<code>onmouseover</code>	Fires when mouse cursor enters element boundary.	<code></code>
<code>onmouseout</code>	Fires when mouse cursor leaves element boundary.	<code></code>
<code>onkeydown</code>	Fires when user depresses a key on keyboard.	<code><input onkeydown="trackKey(event)"></code>
<code>onkeyup</code>	Fires when user releases a key on keyboard.	<code><input onkeyup="validateLength()"></code>
<code>onchange</code>	Fires when input element value is committed/changed (e.g. dropdown pick, checkbox toggle).	<code><select onchange="updateBranch(this.value)"></code>
<code>oninput</code>	Fires instantly as user types or modifies value inside input/textarea.	<code><input oninput="updateLivePreview(this.value)"></code>
<code>onsubmit</code>	Fires when a form submission request is initiated.	<code><form onsubmit="return validateForm()"></code>
<code>onload</code>	Fires when element or document finishes loading completely.	<code><body onload="initApp()"></code>

Why Modern Web Development Prefers JavaScript Event Listeners

While inline event attributes (`onclick=""`) are simple for small beginner examples, modern engineering standards strongly discourage inline event handlers in favor of JavaScript `addEventListener()` separation for 4 major reasons:

1. **Separation of Concerns (SoC):** HTML handles structural markup, CSS handles presentation, and JavaScript handles behavior. Mixing JavaScript into HTML tags creates messy, unmaintainable code.
2. **Security & Content Security Policy (CSP):** Inline event handlers violate strict CSP rules (`unsafe-inline`). Modern secure web applications block inline scripts to prevent Cross-Site Scripting (XSS) attacks.
3. **Multiple Event Binding:** Inline attributes allow only ONE handler string assignment per event type (`onclick="func1()"` overwrites previous inline scripts). `addEventListener()` allows attaching multiple independent listener functions to the same element.
4. **Dynamic DOM Lifecycle:** Elements created dynamically via JavaScript cannot easily bind inline attributes cleanly.

```
<!-- DISCOURAGED (Inline Event Handler) -->
<button onclick="submitData()">Submit</button>

<!-- MODERN BEST PRACTICE (Event Listener in JavaScript File) -->
<button id="submitBtn">Submit</button>

<script>
  document.getElementById("submitBtn").addEventListener("click", function(event) {
    // Clean, secure, modular event handler logic
    console.log("Form Submitted Safely");
  });
</script>
```

3. Void / Empty HTML Elements

Void Elements Reference Table

Void Element	Purpose	Typical Syntax
<code><area></code>	Defines clickable hot-spot area inside an image-map (<code><map></code>).	<code><area shape="rect" coords="0,0,82,126" href="sun.html" alt="Sun"></code>
<code><base></code>	Sets base URL prefix and default target context for relative links.	<code><base href="https://univ.edu/assets/"></code>
<code>
</code>	Inserts a single text line break (newline).	<code><p>Line 1
Line 2</p></code>
<code><col></code>	Sets column properties inside a <code><colgroup></code> table group.	<code><col style="background-color: lightgrey;"></code>
<code><embed></code>	Container for external non-HTML plugin content (Flash/PDF).	<code><embed type="application/pdf" src="doc.pdf" width="300" height="200"></code>
<code><hr></code>	Represents a thematic horizontal rule divider break.	<code><hr></code>
<code></code>	Embeds an image file resource.	<code></code>
<code><input></code>	Interactive form input control.	<code><input type="text" name="usn" placeholder="USN"></code>
<code><link></code>	Specifies resource relationship link (CSS, Favicon).	<code><link rel="stylesheet" href="style.css"></code>
<code><meta></code>	Document metadata configuration.	<code><meta charset="UTF-8"></code>
<code><param></code>	Defines parameters for legacy <code><object></code> plugin elements.	<code><param name="autoplay" value="true"></code>
<code><source></code>	Media source alternative for <code><picture></code> , <code><video></code> , <code><audio></code> .	<code><source src="video.mp4" type="video/mp4"></code>
<code><track></code>	Timed text track subtitle file for video/audio.	<code><track kind="subtitles" src="sub.vtt" srclang="en"></code>
<code><wbr></code>	Word break opportunity for responsive long strings.	<code><p>http://long<wbr>.url</p></code>

What Makes a Void Element Different from a Normal Container Element?

1. **No Closing Tag Allowed:** Normal HTML elements are paired container elements (`<tag>Content</tag>`) that encapsulate text or child tags inside them. Void elements MUST NOT have a closing tag (`</input>` or `</img>` are invalid syntax error in HTML5).

2. **No Inner HTML Content:** Void elements cannot contain nested text or child elements. They are self-contained tags whose behavior is defined entirely through their HTML attributes (e.g. `src` , `type` , `href` , `value`).
3. **Self-Closing Slash Syntax:** In HTML5 syntax standard, trailing self-closing slashes (`<br />` or `<img />`) are completely optional and ignored by HTML5 parsers. `<br>` and `<br />` behave identically.

4. Block-Level vs Inline Elements

In HTML layout rendering, elements traditionally fall into two primary display categories: **Block-Level Elements** and **Inline Elements**.

Detailed Comparison Table

Aspect / Feature	Block-Level Elements	Inline Elements
Default Display	<code>display: block;</code>	<code>display: inline;</code>
Line Break Behavior	Always starts on a new line ; forces subsequent elements to start on a new line.	Fits inline within surrounding text; does NOT break line.
Width Behavior	Takes up 100% full width of parent container by default.	Occupies ONLY as much width as its content requires .
Margin & Padding	Respects all 4 sides (Top, Bottom, Left, Right) of margins & padding.	Respects Left & Right margins/padding; ignores Top & Bottom vertical height margins.
Height & Width CSS	<code>width</code> and <code>height</code> CSS properties can be explicitly set.	<code>width</code> and <code>height</code> CSS properties have no effect .
Child Nesting Rules	Can contain inline elements and OTHER block-level elements.	Can contain ONLY other inline elements (cannot nest block elements inside).
Key Tag Examples	<code><div></code> , <code><p></code> , <code><h1>-<h6></code> , <code><section></code> , <code><article></code> , <code></code> , <code></code> , <code><table></code> , <code><form></code>	<code></code> , <code><a></code> , <code></code> , <code></code> , <code></code> , <code><code></code> , <code><label></code> , <code><input></code> , <code><mark></code>

5. Semantic vs Non-Semantic Elements

HTML5 introduced semantic tags to give meaningful context to web page elements, making web pages readable by both humans, search engines, and screen reader software.

Comparison Breakdown

1. `<div>` vs `<section>`
- `<div>` **(Non-Semantic):** A generic block wrapper container. Carries zero semantic meaning. Used purely as a visual hook for CSS styling or JavaScript DOM wrappers.

- **<section> (Semantic):** Represents a distinct thematic grouping of content, typically with its own heading. Signals to search engines that the enclosed content forms a logical chapter/topic of the document.

2. ** vs Semantic Inline Elements (<time> , <mark> , <code>)**

- ** (Non-Semantic):** A generic inline text container. Carries no semantic meaning. Used for applying inline CSS text styling (color, font).
- **Semantic Inline Elements:** Express specific meaning— **<time>** signals machine dates, **<mark>** signals search highlight relevance, and **<code>** signals computer program code.

3. ** vs **

- ** (Stylistic Bold):** Formats text in bold font weight purely for visual aesthetic offset without implying any extra importance or urgency.
- ** (Semantic Importance):** Formats text in bold font AND signals to screen readers and search engines that the enclosed text has **strong urgency or critical importance**.

4. **<i> vs **

- **<i> (Alternate Voice):** Formats text in italics for foreign words, technical terms, or idiomatic phrases without stress emphasis.
 - ** (Stress Emphasis):** Formats text in italics AND instructs screen readers to pronounce the word with elevated verbal stress emphasis, altering sentence meaning.
-

6. Commonly Used HTML Tags — Quick Reference Table

Tag Name	Primary Function / Purpose	Key HTML Attributes
<code><html></code>	Document root element container.	<code>lang</code>
<code><head></code>	Document metadata container.	Global attributes
<code><title></code>	Browser tab title label.	Global attributes
<code><body></code>	Main renderable document content.	Global attributes
<code><meta></code>	Character encoding, viewport, SEO settings.	<code>charset</code> , <code>name</code> , <code>content</code> , <code>http-equiv</code>
<code><link></code>	Attach CSS stylesheets / fonts	<code>rel</code> , <code>href</code> , <code>type</code> , <code>media</code>
<code><h1>-<h6></code>	Document headings (Level 1 to Level 6).	<code>id</code> , <code>class</code>
<code><p></code>	Text paragraph block.	Global attributes
<code><a></code>	Hyperlink anchor element.	<code>href</code> , <code>target</code> , <code>rel</code> , <code>download</code>
<code></code>	Image embedding tag.	<code>src</code> , <code>alt</code> , <code>width</code> , <code>height</code> , <code>loading</code>
<code></code>	Unordered bulleted list.	Global attributes
<code></code>	Sequenced numbered list.	<code>type</code> , <code>start</code> , <code>reversed</code>
<code></code>	List item entry inside lists.	<code>value</code> (in ol)
<code><table></code>	Matrix data grid table.	<code>border</code>
<code><tr></code>	Table row container.	Global attributes
<code><th></code>	Table header cell (bold, centered).	<code>scope</code> , <code>colspan</code> , <code>rowspan</code>
<code><td></code>	Table data cell.	<code>colspan</code> , <code>rowspan</code>
<code><form></code>	User input form container.	<code>action</code> , <code>method</code> , <code>enctype</code> , <code>autocomplete</code>
<code><input></code>	Interactive form input control.	<code>type</code> , <code>name</code> , <code>value</code> , <code>placeholder</code> , <code>required</code>
<code><label></code>	Accessible caption for input fields.	<code>for</code>
<code><button></code>	Clickable button control.	<code>type</code> , <code>disabled</code>
<code><textarea></code>	Multi-line text input field.	<code>name</code> , <code>rows</code> , <code>cols</code> , <code>placeholder</code>
<code><select></code>	Dropdown select menu container.	<code>name</code> , <code>multiple</code> , <code>required</code>
<code><option></code>	Dropdown selection item choice.	<code>value</code> , <code>selected</code> , <code>disabled</code>
<code><header></code>	Semantic top header section.	Global attributes
<code><footer></code>	Semantic bottom footer section.	Global attributes

Tag Name	Primary Function / Purpose	Key HTML Attributes
<code><main></code>	Unique primary page content landmark.	Global attributes
<code><section></code>	Thematic document chapter section.	<code>id</code>
<code><article></code>	Standalone reusable article post.	Global attributes
<code><aside></code>	Sidebar or auxiliary callout content.	Global attributes
<code><div></code>	Generic non-semantic block container.	<code>id</code> , <code>class</code> , <code>style</code>
<code></code>	Generic non-semantic inline container.	<code>class</code> , <code>style</code>
<code><details></code>	Interactive accordion disclosure box.	<code>open</code>
<code><summary></code>	Heading label for <code><details></code> .	Global attributes
<code><script></code>	JavaScript code embedding tag.	<code>src</code> , <code>async</code> , <code>defer</code> , <code>type</code>

7. Complete HTML5 Webpage Example

Below is a complete, fully functional HTML5 webpage demonstrating standard document structure, semantic layout tags, headings, text formatting, media elements, tables, forms, and interactive widgets.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <!-- Metadata Configuration -->
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description" content="Complete College Web Technology Portal Demonstration">
  <meta name="author" content="Aryansh Sharma">
  <title>VTU CSE Department Portal | Web Tech Lab Assignment</title>

  <!-- Internal CSS Styling -->
  <style>
    body {
      font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
      margin: 0;
      padding: 0;
      background-color: #f8fafc;
      color: #1e293b;
      line-height: 1.6;
    }
    header {
      background-color: #0f172a;
      color: white;
      padding: 1.5rem;
      text-align: center;
    }
    nav {
      background-color: #1e293b;
      padding: 0.75rem;
      text-align: center;
    }
    nav a {
      color: #38bdf8;
      margin: 0 15px;
      text-decoration: none;
      font-weight: bold;
    }
    nav a:hover {
      text-decoration: underline;
    }
    .container {
      max-width: 1100px;
      margin: 20px auto;
      padding: 0 20px;
      display: flex;
      gap: 20px;
    }
    main {
      flex: 3;
      background: white;
      padding: 25px;
      border-radius: 8px;
      box-shadow: 0 1px 3px rgba(0,0,0,0.1);
    }
    aside {
      flex: 1;
```

```
        background: #f1f5f9;
        padding: 20px;
        border-radius: 8px;
        height: fit-content;
    }
    table {
        width: 100%;
        border-collapse: collapse;
        margin: 15px 0;
    }
    table, th, td {
        border: 1px solid #cbd5e1;
    }
    th, td {
        padding: 10px;
        text-align: left;
    }
    th {
        background-color: #e2e8f0;
    }
    fieldset {
        border: 1px solid #cbd5e1;
        border-radius: 6px;
        padding: 15px;
        margin-bottom: 15px;
    }
    legend {
        font-weight: bold;
        color: #0f172a;
    }
    footer {
        background-color: #0f172a;
        color: white;
        text-align: center;
        padding: 1rem;
        margin-top: 30px;
    }
</style>
</head>
<body>

<!-- Document Header -->
<header>
    <h1>Visvesvaraya Technological University</h1>
    <p>Department of Computer Science & Engineering</p>
</header>

<!-- Primary Navigation -->
<nav aria-label="Main Menu">
    <a href="#about">About Course</a>
    <a href="#syllabus">Syllabus</a>
    <a href="#registration">Registration</a>
    <a href="#faq">FAQ</a>
</nav>

<!-- Main Layout Container -->
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<div class="container">

  <!-- Primary Content Area -->
  <main>

    <!-- About Section -->
    <section id="about">
      <article>
        <h2>Course Overview: Web Technologies (18CS52)</h2>
        <p>This course introduces engineering students to fundamental web technologies includi
        <p>Key highlights include building <strong>semantic accessible web structures</strong>
      </article>
    </section>

    <hr>

    <!-- Syllabus Table Section -->
    <section id="syllabus">
      <h3>Semester 3 Examination Timetable</h3>
      <table>
        <caption>Table 1: CSE Exam Schedule - 2026</caption>
        <thead>
          <tr>
            <th scope="col">Subject Code</th>
            <th scope="col">Subject Title</th>
            <th scope="col">Exam Date</th>
            <th scope="col">Session</th>
          </tr>
        </thead>
        <tbody>
          <tr>
            <td>18CS51</td>
            <td>Management and Entrepreneurship</td>
            <td><time datetime="2026-09-01">01 Sept 2026</time></td>
            <td>Morning</td>
          </tr>
          <tr>
            <td>18CS52</td>
            <td>Web Technology & Its Applications</td>
            <td><time datetime="2026-09-04">04 Sept 2026</time></td>
            <td>Morning</td>
          </tr>
        </tbody>
        <tfoot>
          <tr>
            <td colspan="4">Note: Hall tickets must be verified prior to entry.</td>
          </tr>
        </tfoot>
      </table>
    </section>

    <hr>

    <!-- Interactive Registration Form Section -->
    <section id="registration">
      <h3>Student Lab Assignment Submission Form</h3>

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<form action="/submit-lab" method="POST" enctype="multipart/form-data">

  <fieldset>
    <legend>Student Information</legend>

    <p>
      <label for="usn">USN Number:</label><br>
      <input type="text" id="usn" name="usn_number" placeholder="e.g. 1VU22CS001" required />
    </p>

    <p>
      <label for="email">College Email:</label><br>
      <input type="email" id="email" name="student_email" placeholder="student@college.edu" />
    </p>

    <p>
      <label for="branch">Engineering Branch:</label><br>
      <select id="branch" name="branch" required>
        <option value="" disabled selected>-- Choose Branch --</option>
        <option value="cse">Computer Science (CSE)</option>
        <option value="ise">Information Science (ISE)</option>
        <option value="ece">Electronics (ECE)</option>
      </select>
    </p>
  </fieldset>

  <fieldset>
    <legend>Assignment File Upload</legend>
    <p>
      <label for="assignment">Upload Code File (.zip, .pdf):</label><br>
      <input type="file" id="assignment" name="assignment_file" accept=".zip,.pdf" required />
    </p>
    <p>
      <label for="comments">Student Remarks:</label><br>
      <textarea id="comments" name="remarks" rows="3" placeholder="Enter comments here.." />
    </p>
  </fieldset>

  <button type="submit">Submit Assignment</button>
  <button type="reset">Reset Form</button>
</form>
</section>

<hr>

<!-- Interactive Details Widget -->
<section id="faq">
  <h3>Frequently Asked Questions</h3>
  <details>
    <summary>What is the submission deadline?</summary>
    <p>All HTML assignments must be submitted before <time datetime="2026-08-31T23:59">Aug 31, 2026</time>.</p>
  </details>
  <details>
    <summary>Can I resubmit my assignment?</summary>
    <p>Yes, multiple submissions are permitted prior to the final cutoff date.</p>
  </details>

```

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    </section>

</main>

<!-- Sidebar Auxiliary Area -->
<aside>
  <h3>Important Announcements</h3>
  <ul>
    <li><mark>New</mark>: Web Tech Lab Viva starts next week.</li>
    <li>Download official <a href="#syllabus" download>Syllabus Copy (PDF)</a>.</li>
  </ul>

  <hr>

  <h3>Lab Location</h3>
  <address>
    Computing Lab 3, CSE Block<br>
    VTU Campus, Belagavi<br>
    Karnataka - 590018
  </address>
</aside>

</div>

<!-- Document Footer -->
<footer>
  <p>&copy; 2026 Visvesvaraya Technological University. All Rights Reserved.</p>
  <p><small>Designed for Computer Science Engineering Web Development Curriculum.</small></p>
</footer>

<!-- External JavaScript -->
<script>
  console.log("VTU Portal Loaded Successfully.");
</script>
</body>
</html>

```

Code Walkthrough Section-by-Section

- Document Header & Metadata** (`<!DOCTYPE html>` , `<html>` , `<head>` , `<meta>` , `<title>`): Sets HTML5 standard parsing, defines UTF-8 character encoding, responsive viewport sizing for mobile devices, page title bar text, and embedded CSS rules.
- Structural Header & Navigation** (`<header>` , `<nav>` , `<a>`): Encapsulates top site branding and provides accessible navigation hyperlinking to internal page sections (`#about` , `#syllabus` , `#registration`).
- Main & Sidebar Container** (`<div class="container">` , `<main>` , `<aside>`): Creates a responsive 2-column flexbox layout separating primary content (`<main>`) from secondary news/announcements (`<aside>`).
- Article & Sectioning** (`<section>` , `<article>` , `<h2>` , `<p>` , `<abbr>`): Organizes course overview content into semantic sections, using `<abbr>` for acronym tooltips (HTML5, CSS3).
- Tabular Schedule Data** (`<table>` , `<caption>` , `<thead>` , `<tbody>` , `<tfoot>` , `<th>` , `<td>` , `<time>`): Presents exam schedule with proper column header scopes (`scope="col"`), table caption,

footer notes, and ISO date markup (`<time>`).

6. **Form Processing** (`<form>` , `<fieldset>` , `<legend>` , `<label>` , `<input>` , `<select>` , `<textarea>` , `<button>`): Implements student registration form with input pattern validation, file upload support (`enctype="multipart/form-data"`), dropdown choices, multi-line comment textareas, and submit buttons.
7. **Interactive Elements** (`<details>` , `<summary>`): Implements accordion FAQ dropdowns that expand/collapse natively without requiring JavaScript code.
8. **Sidebar Contact Information** (`<aside>` , `<address>` , `<mark>` , `<small>`): Uses `<address>` for physical contact details, `<mark>` for highlighted announcement badges, and `<small>` for footer legal fine print.

8. HTML Tag Cheat Sheet

Tag → Purpose → Key Attribute(s)

- `<html>` → Root document container → `lang`
- `<head>` → Document metadata wrapper → Global attributes
- `<title>` → Browser tab title text → Global attributes
- `<body>` → Visible document content canvas → Global attributes
- `<meta>` → Viewport, charset, SEO metadata → `charset` , `name` , `content`
- `<link>` → Attach CSS stylesheets / fonts → `rel` , `href` , `type`
- `<style>` → Embedded internal CSS block → `media`
- `<h1>--<h6>` → Heading hierarchy → `id` , `class`
- `<p>` → Paragraph text block → Global attributes
- `<br>` → Line break (newline) → None (Void element)
- `<hr>` → Horizontal thematic rule divider → None (Void element)
- `<pre>` → Preformatted monospace text → Global attributes
- `<blockquote>` → Extended indented block quote → `cite`
- `<q>` → Short inline quotation → `cite`
- `<abbr>` → Abbreviation or acronym tooltip → `title`
- `<code>` → Inline program code fragment → Global attributes
- `<kbd>` → Keyboard key entry display → Global attributes
- `<samp>` → Sample program output message → Global attributes
- `<var>` → Mathematical/programming variable → Global attributes
- `<strong>` → Strong importance (bold) → Global attributes
- `<em>` → Stress vocal emphasis (italics) → Global attributes
- `<mark>` → Text highlighter marker → Global attributes
- `<small>` → Legal fine print sub-text → Global attributes
- `<sub>` → Subscript text (H₂O) → Global attributes
- `<sup>` → Superscript text (x²) → Global attributes
- `<del>` → Deleted text (strikethrough) → `cite` , `datetime`
- `<ins>` → Inserted text (underline) → `cite` , `datetime`

- `<a>` → Hyperlink anchor → `href`, `target`, `rel`, `download`
- `<nav>` → Navigation link block → `aria-label`
- `<img>` → Image embed tag → `src`, `alt`, `width`, `height`, `loading`
- `<picture>` → Responsive image container → Global attributes
- `<source>` → Media source format choice → `srcset`, `src`, `type`, `media`
- `<figure>` → Image/diagram wrapper block → Global attributes
- `<figcaption>` → Figure title caption → Global attributes
- `<audio>` → Embedded audio player → `src`, `controls`, `autoplay`, `loop`
- `<video>` → Embedded video player → `src`, `controls`, `poster`, `width`, `height`
- `<track>` → Subtitles / CC track file → `kind`, `src`, `srclang`, `label`
- `<iframe>` → Embedded inline webpage window → `src`, `width`, `height`, `title`, `sandbox`
- `<ul>` → Unordered bulleted list → Global attributes
- `<ol>` → Ordered numbered list → `type`, `start`, `reversed`
- `<li>` → List item entry → `value` (in `<ol>`)
- `<dl>` → Description list wrapper → Global attributes
- `<dt>` → Description list term/key → Global attributes
- `<dd>` → Description list value/definition → Global attributes
- `<table>` → Matrix data grid table → `border`
- `<caption>` → Table title caption → Global attributes
- `<thead>` → Table header row group → Global attributes
- `<tbody>` → Table body data row group → Global attributes
- `<tfoot>` → Table summary footer row group → Global attributes
- `<tr>` → Table row container → Global attributes
- `<th>` → Table header cell (bold) → `scope`, `colspan`, `rowspan`
- `<td>` → Table data cell → `colspan`, `rowspan`
- `<form>` → Interactive form container → `action`, `method`, `enctype`, `autocomplete`
- `<input>` → Interactive input control → `type`, `name`, `value`, `placeholder`, `required`
- `<label>` → Accessible input text label → `for`
- `<button>` → Clickable button element → `type`, `disabled`
- `<textarea>` → Multi-line prose text box → `name`, `rows`, `cols`, `placeholder`
- `<select>` → Dropdown selection menu → `name`, `multiple`, `required`
- `<option>` → Dropdown option choice → `value`, `selected`, `disabled`
- `<optgroup>` → Dropdown category group → `label`, `disabled`
- `<fieldset>` → Form group bordered box → `disabled`
- `<legend>` → Fieldset title caption → Global attributes
- `<datalist>` → Auto-suggest list for inputs → Global attributes (`id`)
- `<output>` → Calculated result display → `for`, `name`
- `<progress>` → Task progress bar gauge → `value`, `max`
- `<meter>` → Scalar range measurement gauge → `value`, `min`, `max`, `low`, `high`, `optimum`
- `<header>` → Page/Section top header → Global attributes
- `<footer>` → Page/Section bottom footer → Global attributes
- `<main>` → Primary page content landmark → Global attributes

- `<section>` → Thematic chapter section → `id`
 - `<article>` → Reusable independent article → Global attributes
 - `<aside>` → Sidebar callout container → Global attributes
 - `<div>` → Generic block wrapper container → `id`, `class`, `style`
 - `<span>` → Generic inline text wrapper container → `class`, `style`
 - `<details>` → Accordion disclosure widget → `open`
 - `<summary>` → Accordion header label → Global attributes
 - `<dialog>` → Native modal window popup → `open`
 - `<script>` → JavaScript execution block → `src`, `async`, `defer`, `type`
 - `<noscript>` → Fallback when JS disabled → Global attributes
 - `<canvas>` → Dynamic 2D/3D graphics canvas → `width`, `height`
 - `<template>` → Hidden reusable HTML template → Global attributes (`id`)
 - `<slot>` → Web Component shadow DOM slot → `name`
 - `<time>` → Machine date/time tag → `datetime`
 - `<data>` → Machine translation value tag → `value`
 - `<ruby>` → East Asian furigana wrapper → Global attributes
 - `<rt>` → East Asian phonetic guide text → Global attributes
 - `<rp>` → Non-ruby fallback parenthesis → Global attributes
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